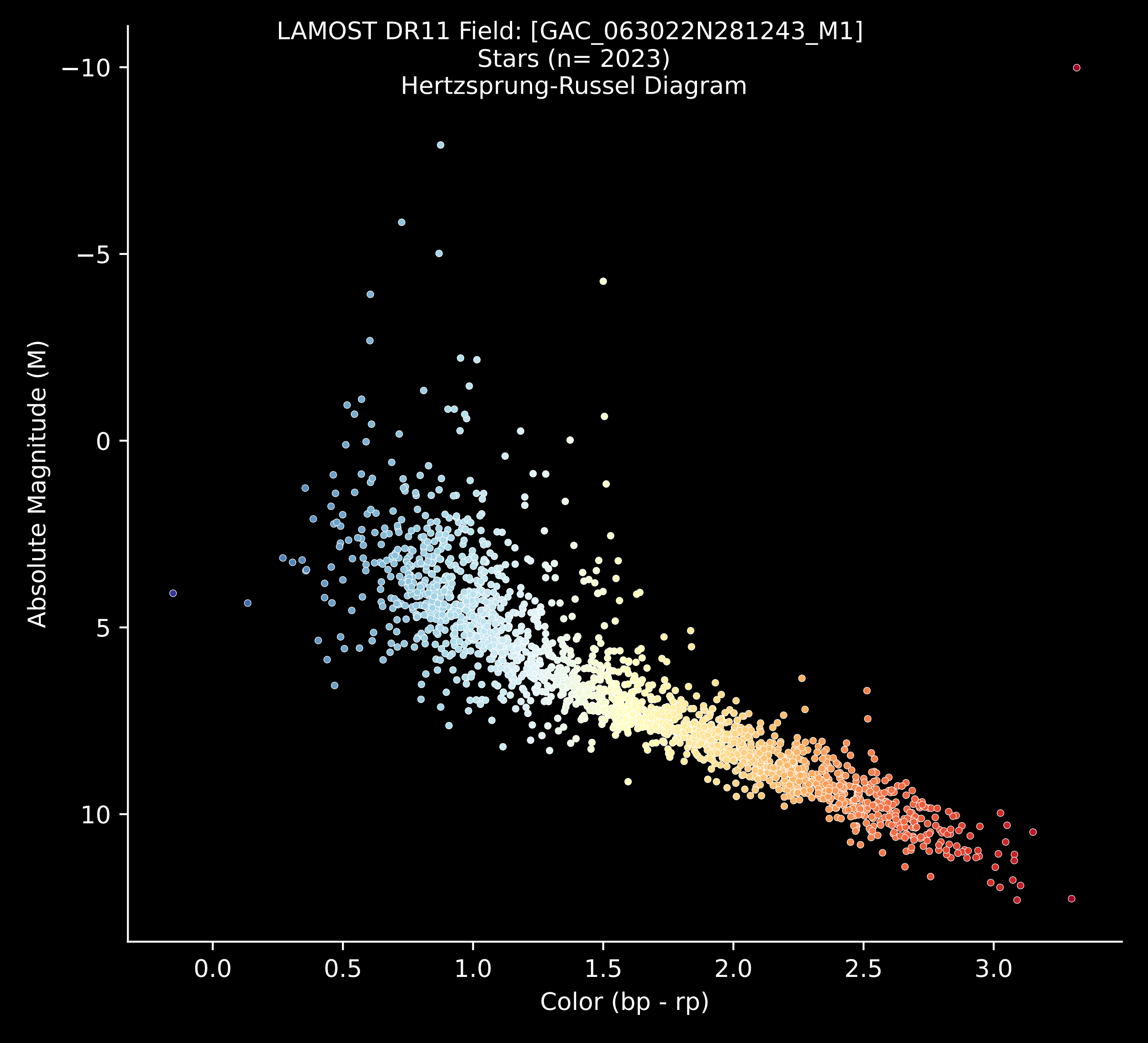
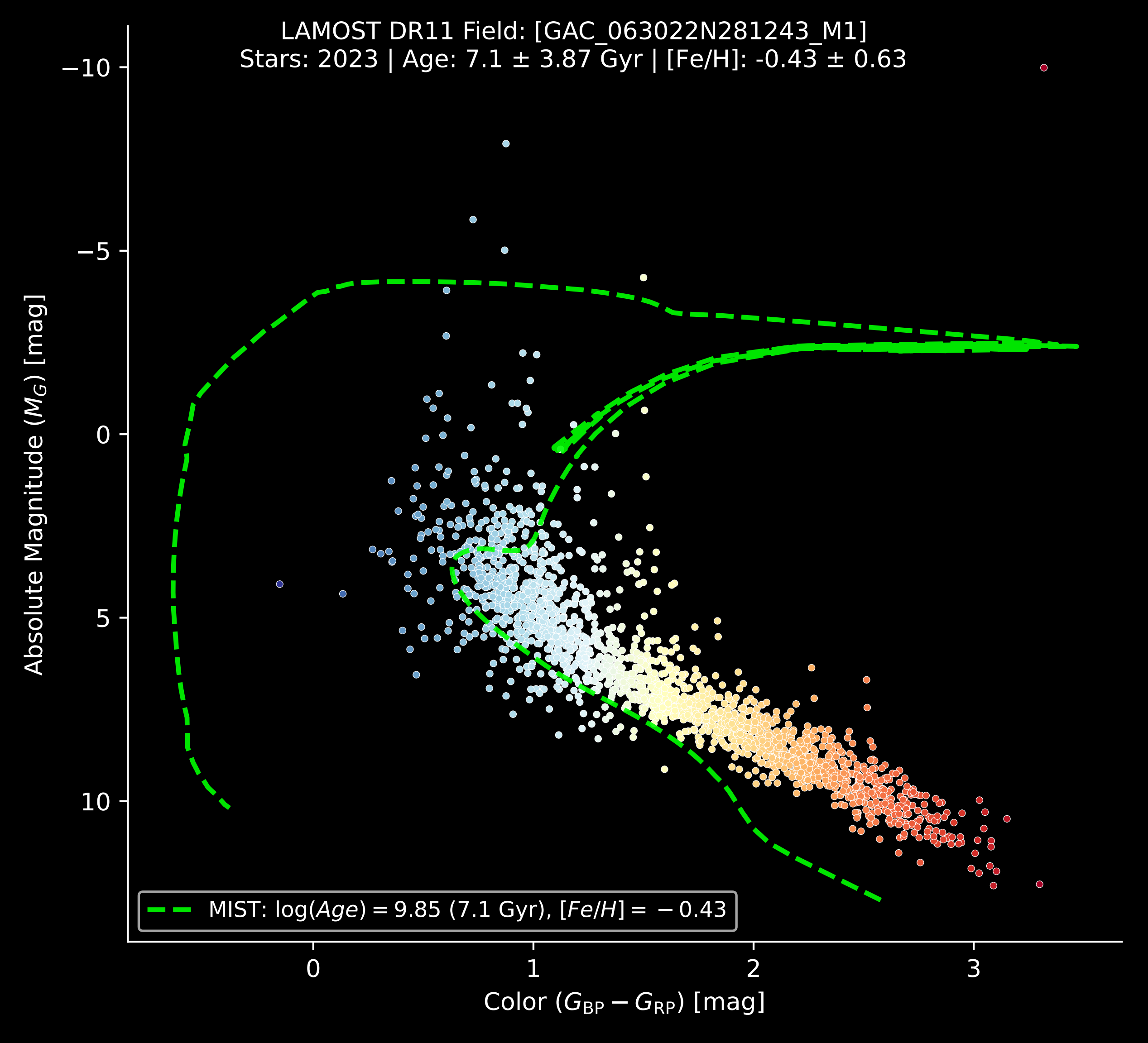
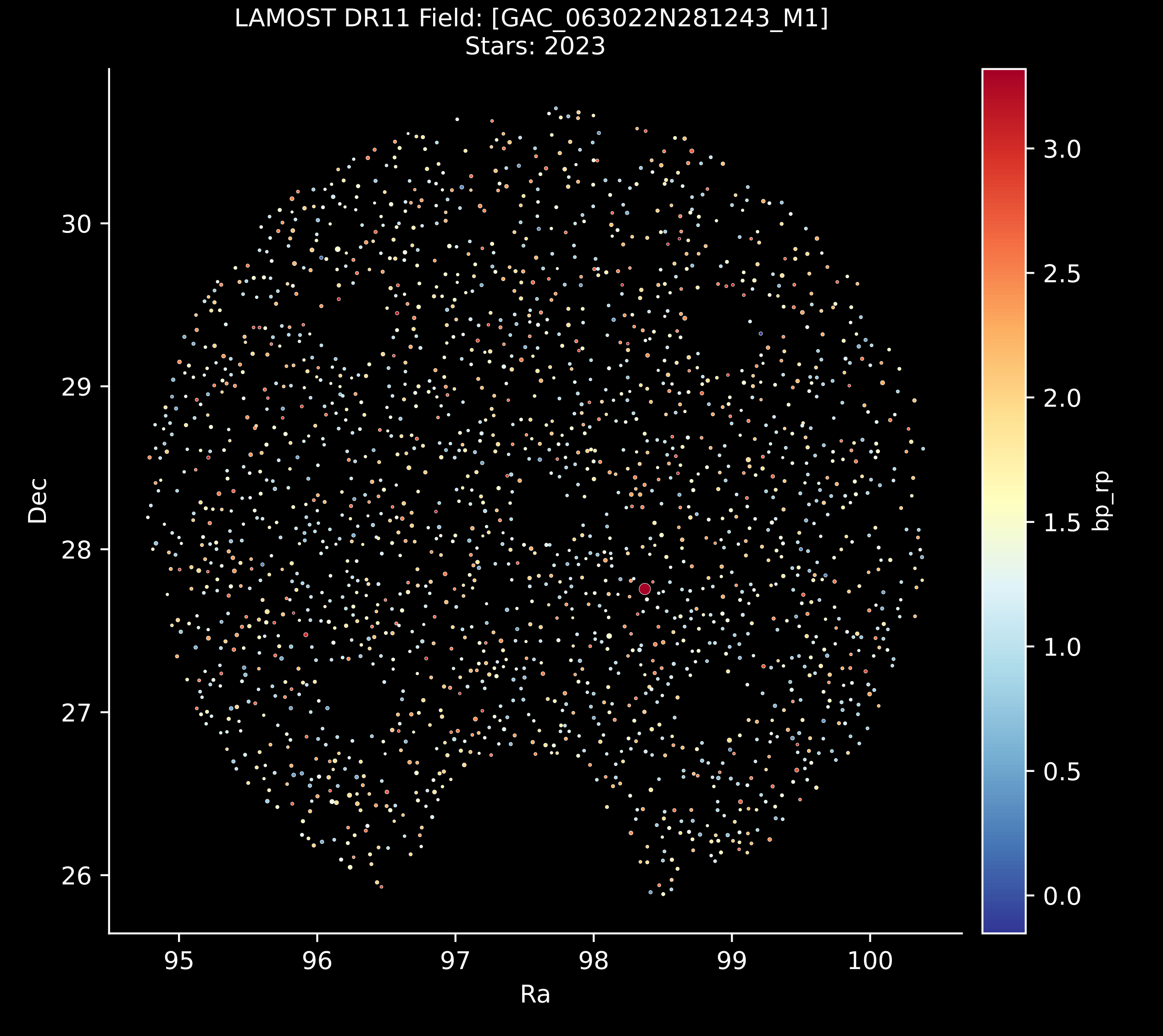
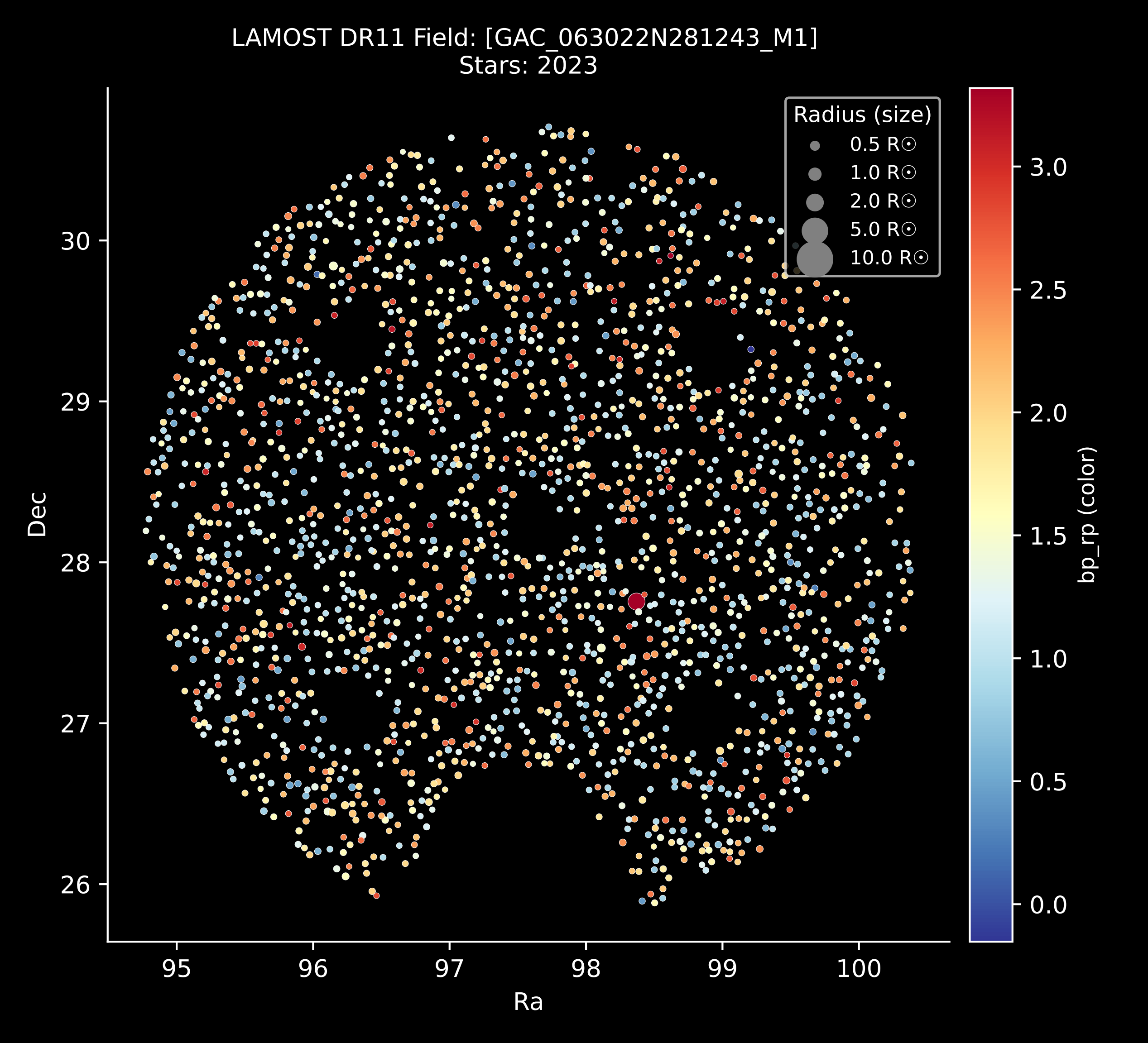

LAMOST DR11 Field: [GAC_063022N281243_M1]
Stars (n= 2023)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−10
−5
0
5
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0

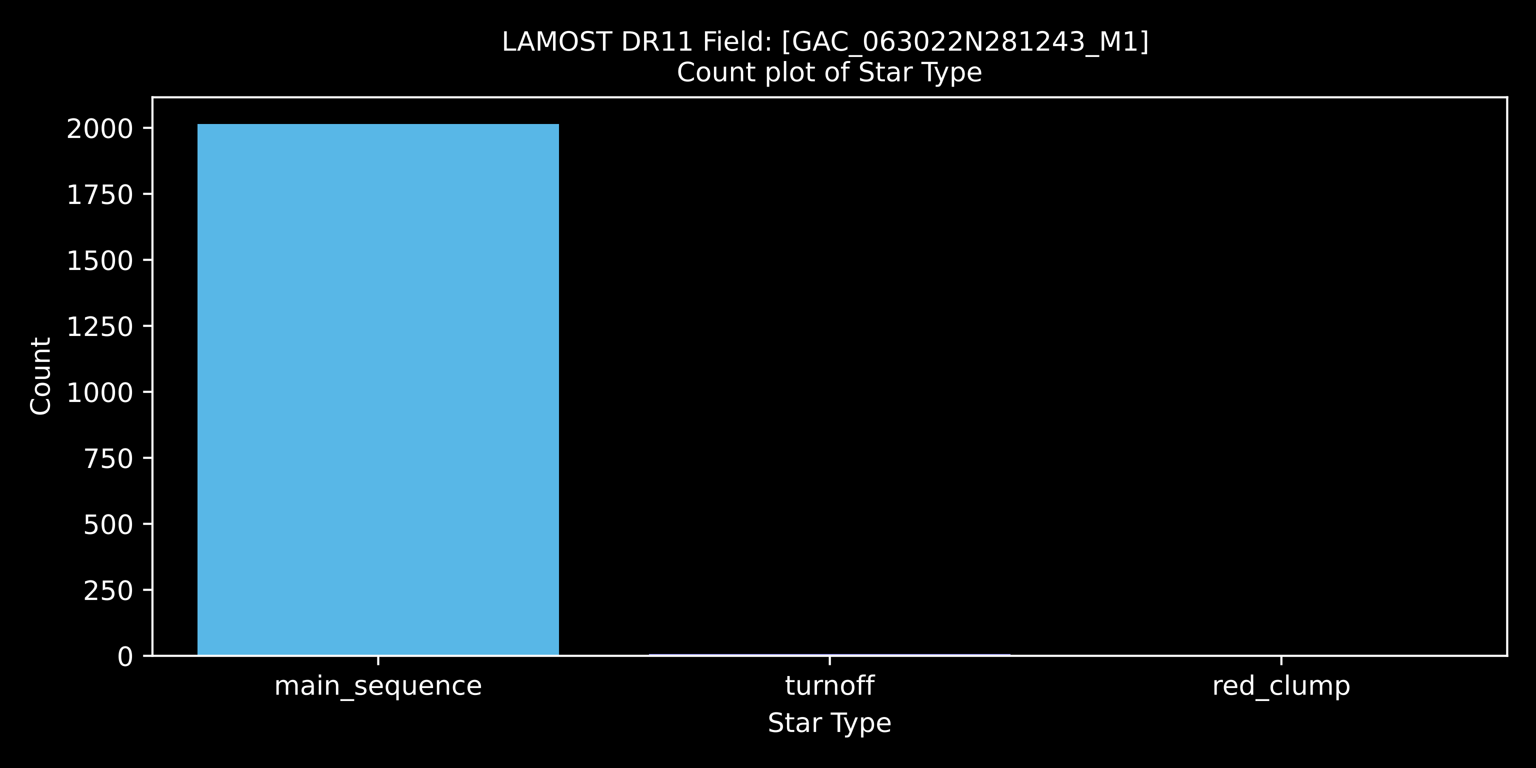

LAMOST DR11 Field: [GAC_063022N281243_M1]
Stars: 2023 | Age: 7.1 ± 3.87 Gyr | [Fe/H]: -0.43 ± 0.63
Absolute Magnitude (M_G) [mag]
Color ($G_{\mathrm{BP}} - G_{\mathrm{RP}}$) [mag]
MIST: $\log(Age) = 9.85$ (7.1 Gyr), $[Fe/H] = -0.43$
−10
−5
0
5
10
0
1
2
3



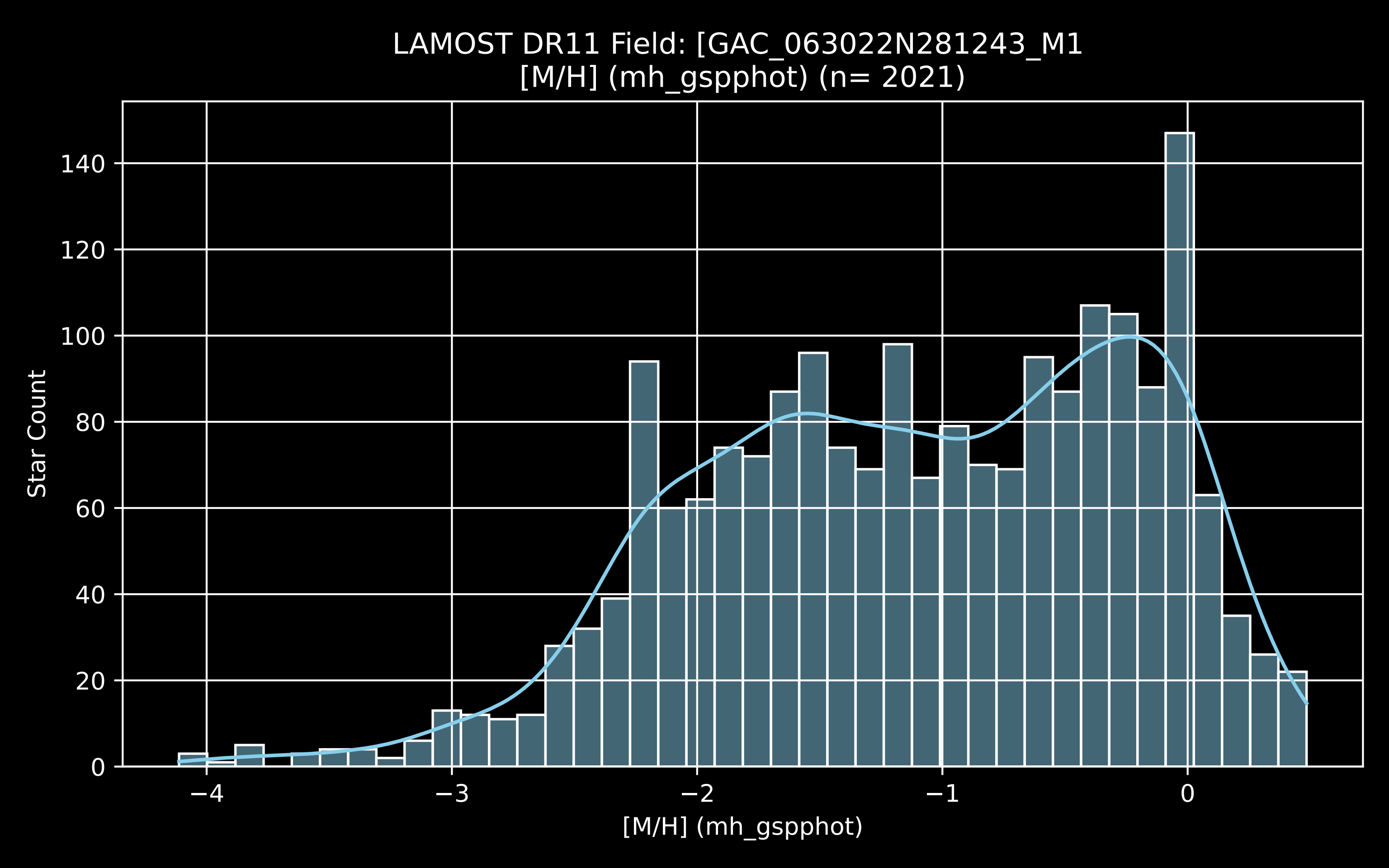
LAMOST DR11 Field: [GAC_063022N281243_M1]
Stars: 2023
Dec
Ra
bp_rp
30
29
28
27
26
95
96
97
98
99
100
3.0
2.5
2.0
1.5
1.0
0.5
0.0



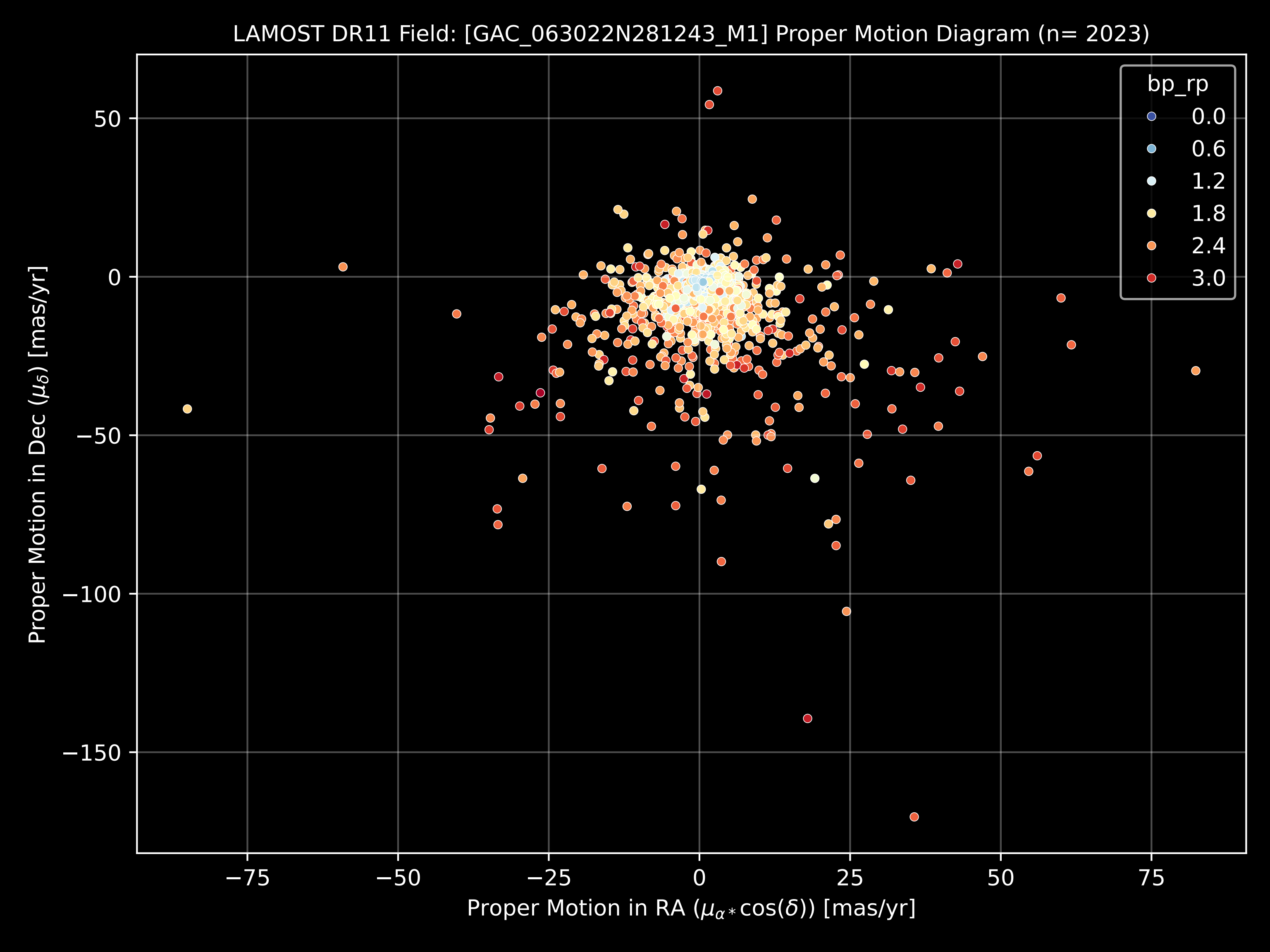
LAMOST DR11 Field: [GAC_063022N281243_M1]
Stars: 2023
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
26
27
28
29
30
95
96
97
98
99
100
0.0
0.5
1.0
1.5
2.0
2.5
3.0



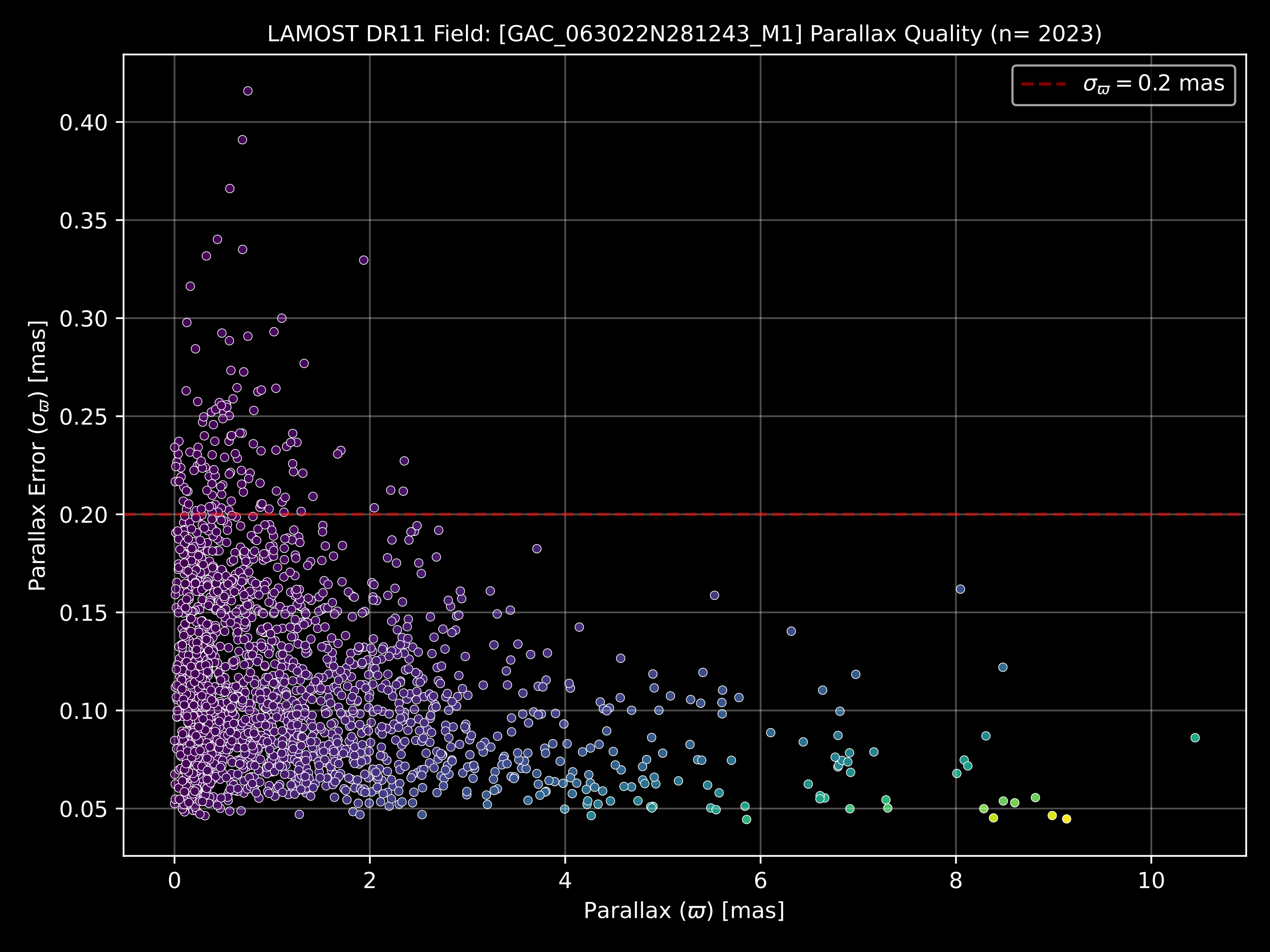
LAMOST DR11 Field: [GAC_063022N281243_M1]
Count plot of Star Type
Count
2000
1750
1500
1250
1000
750
500
250
0
main_sequence
turnoff
red_clump
Star Type

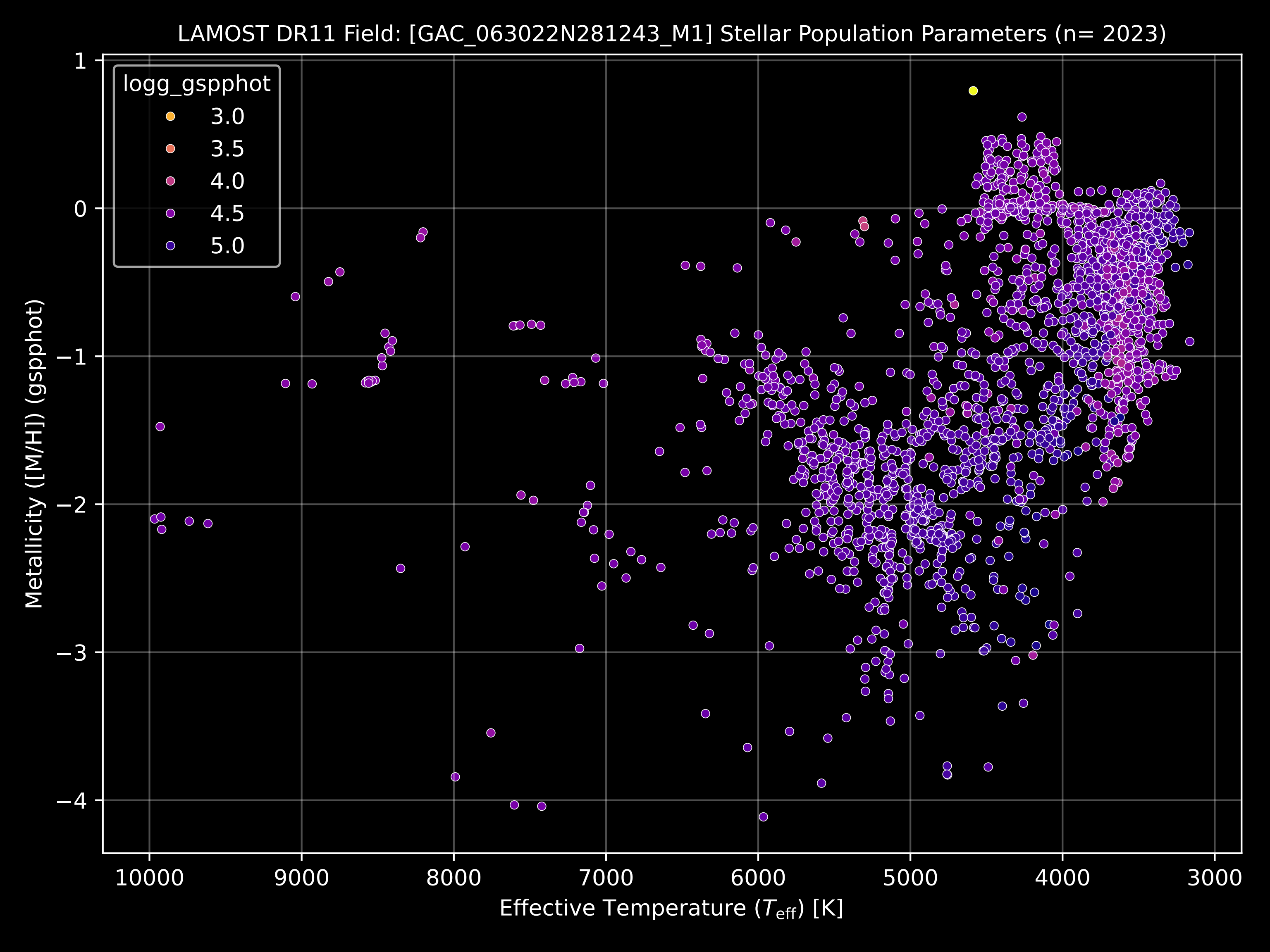


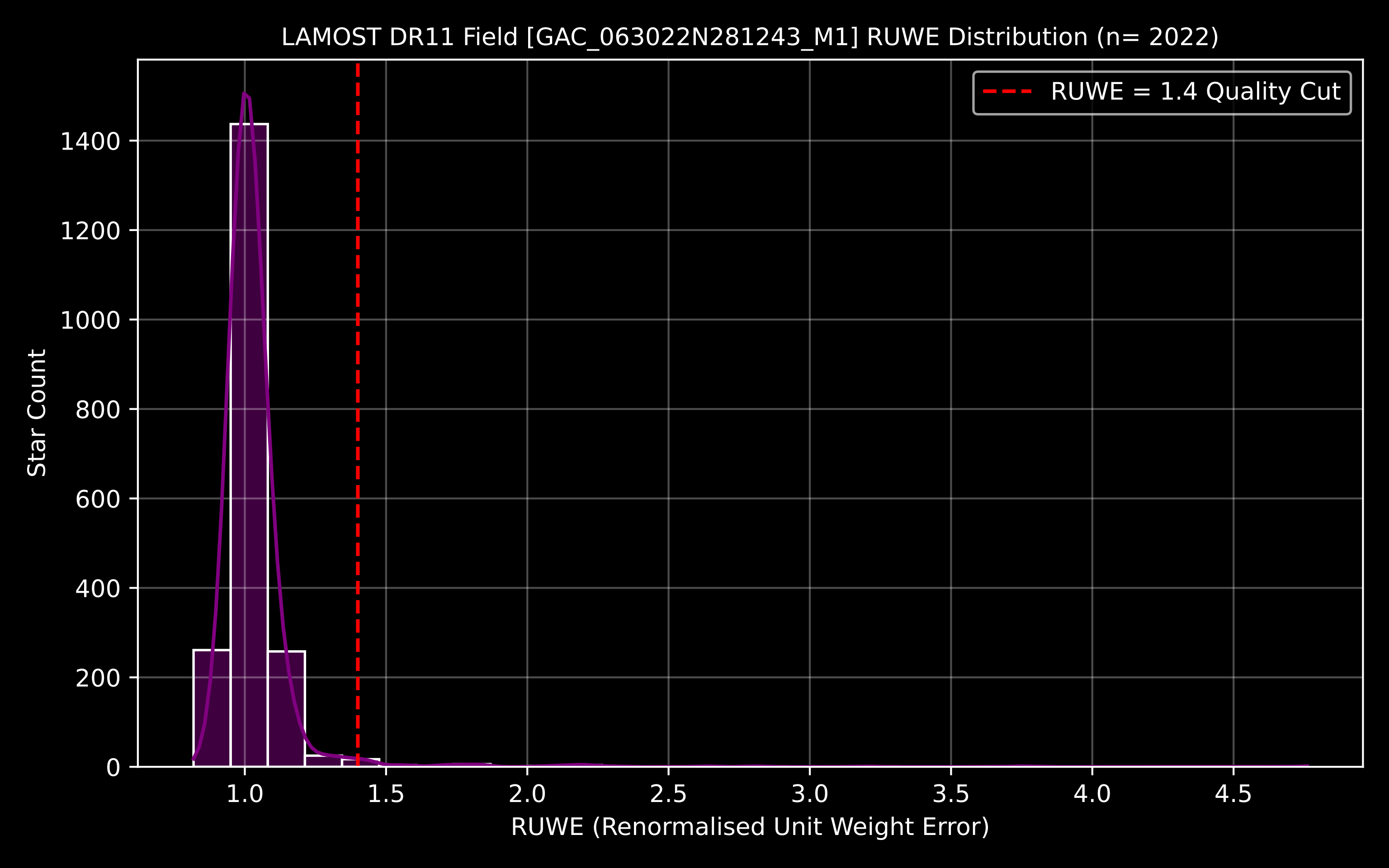
LAMOST DR11 Field: [GAC_063022N281243_M1
[M/H] (mh_gspphot) (n= 2021)
Star Count
[M/H] (mh_gspphot)
0
20
40
60
80
100
120
140
−4
−3
−2
−1
0

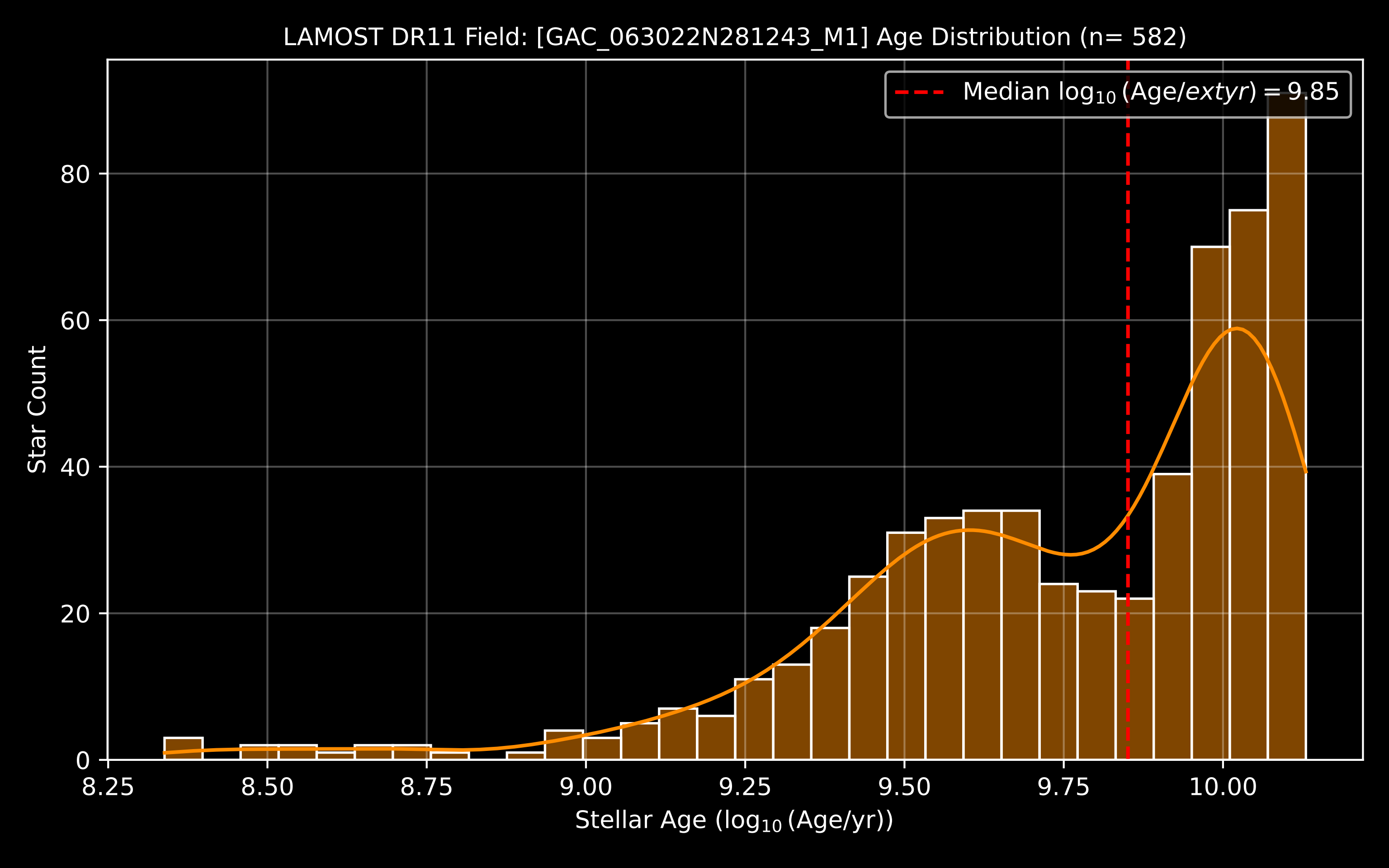


LAMOST DR11 Field: [GAC_063022N281243_M1] Proper Motion Diagram (n= 2023)
bp_rp
0.0
0.6
1.2
1.8
2.4
3.0
50
0
−50
−100
−150
−75
−50
−25
0
25
50
75
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

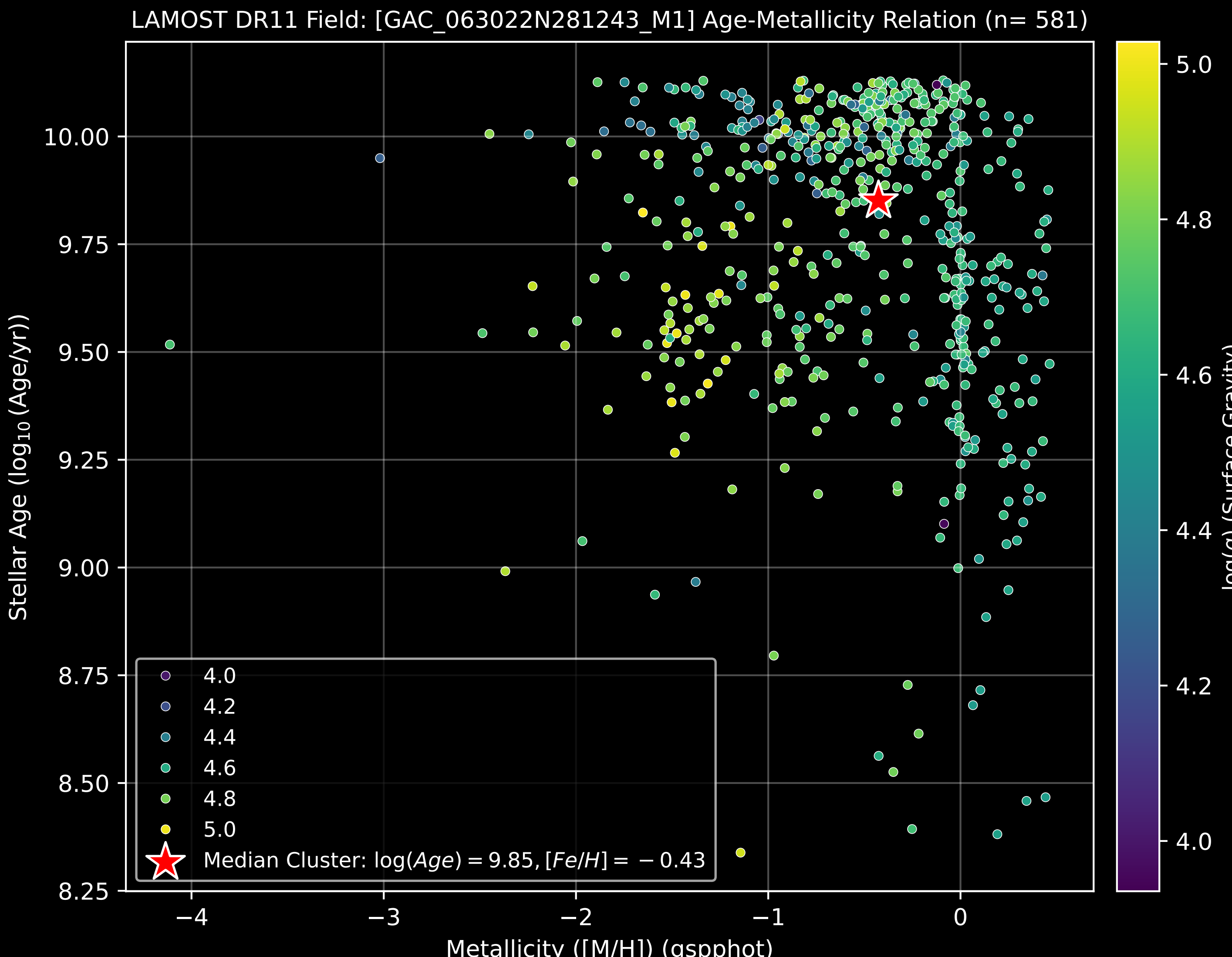

LAMOST DR11 Field: [GAC_063022N281243_M1] Parallax Quality (n= 2023)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
2
4
6
8
10


LAMOST DR11 Field: [GAC_063022N281243_M1] Stellar Population Parameters (n= 2023)
logg_gspphot
3.0
3.5
4.0
4.5
5.0
Metallicity ([M/H]) (gspphot)
Effective Temperature (T_{eff}) [K]
1
0
−1
−2
−3
−4
10000
9000
8000
7000
6000
5000
4000
3000

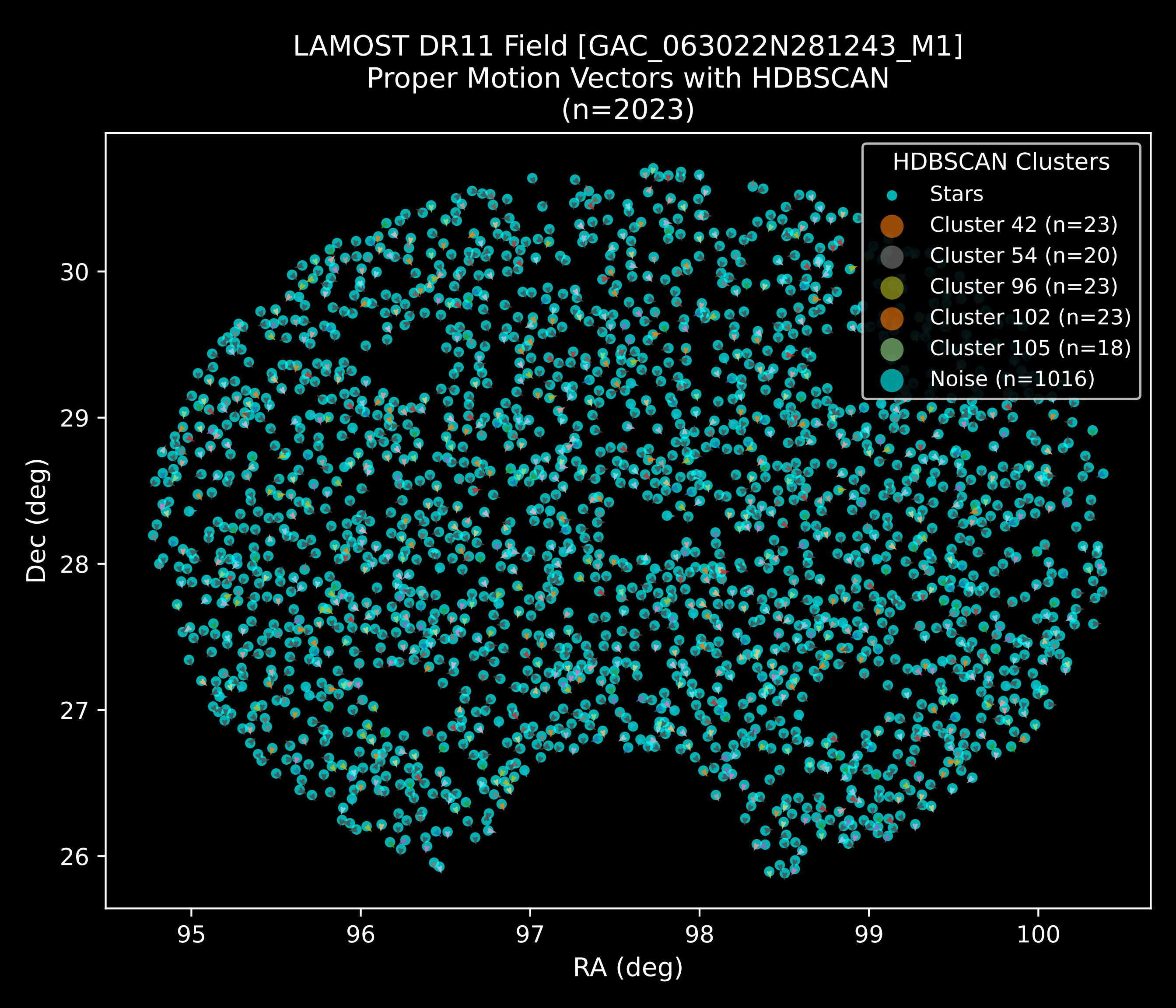

LAMOST DR11 Field [GAC_063022N281243_M1] RUWE Distribution (n= 2022)
RUWE = 1.4 Quality Cut
Star Count
RUWE (Renormalised Unit Weight Error)
0
200
400
600
800
1000
1200
1400
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5

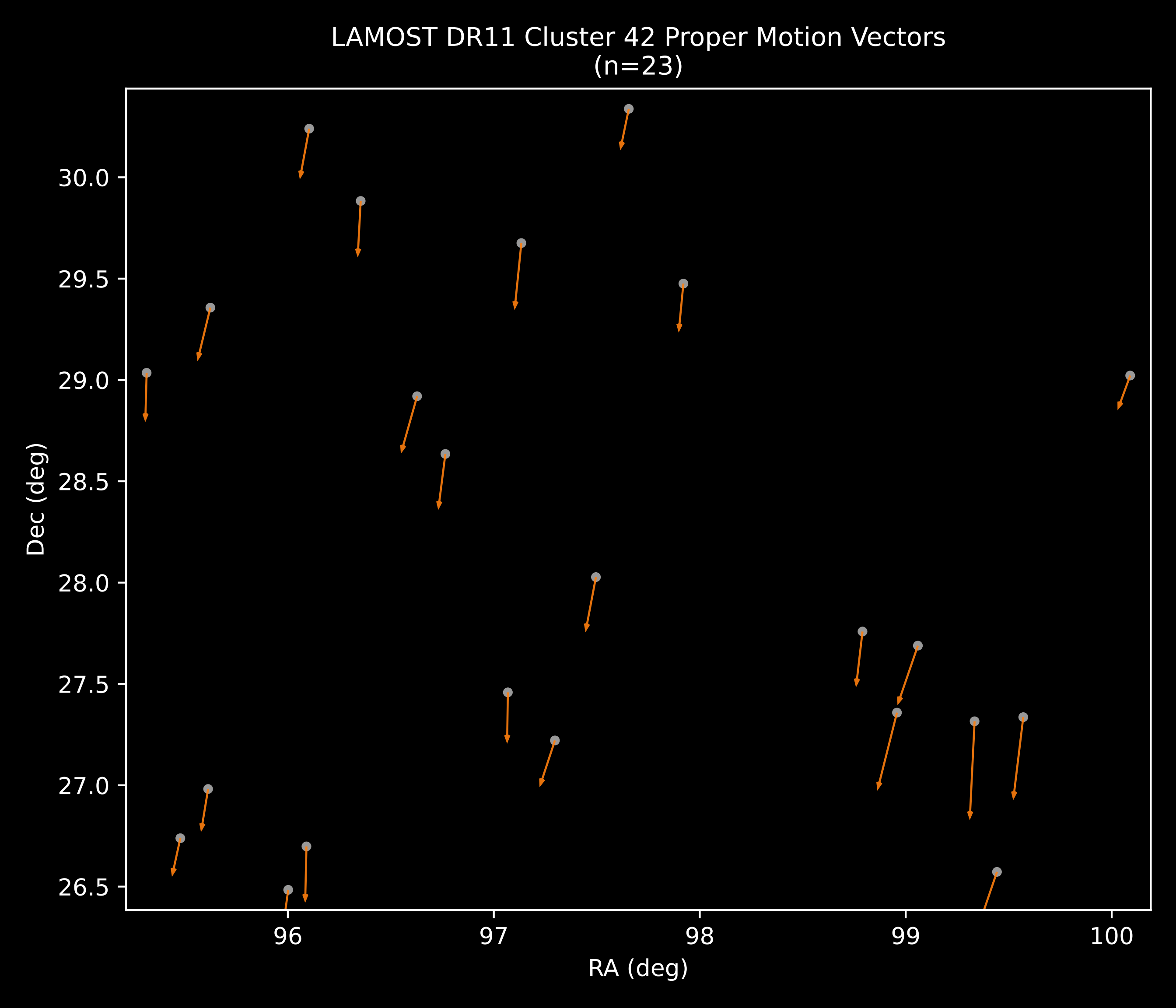


LAMOST DR11 Field: [GAC_063022N281243_M1] Age Distribution (n= 582)
Median $\log_{10}(\mathrm{Age}/extyr) = 9.85$
Star Count
0
20
40
60
80
8.25
8.50
8.75
9.00
9.25
9.50
9.75
10.00
Stellar Age ($\log_{10}$(Age/yr))

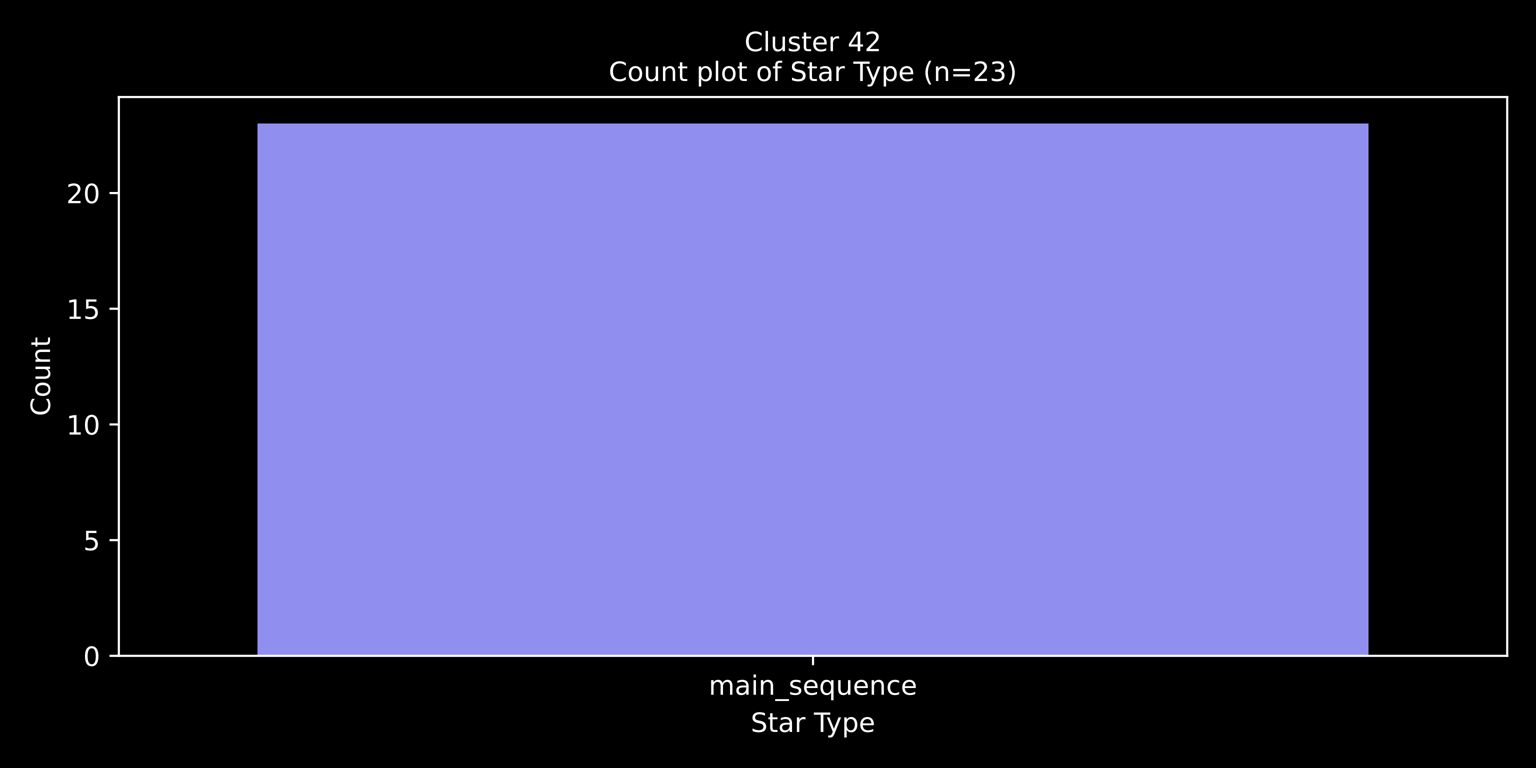


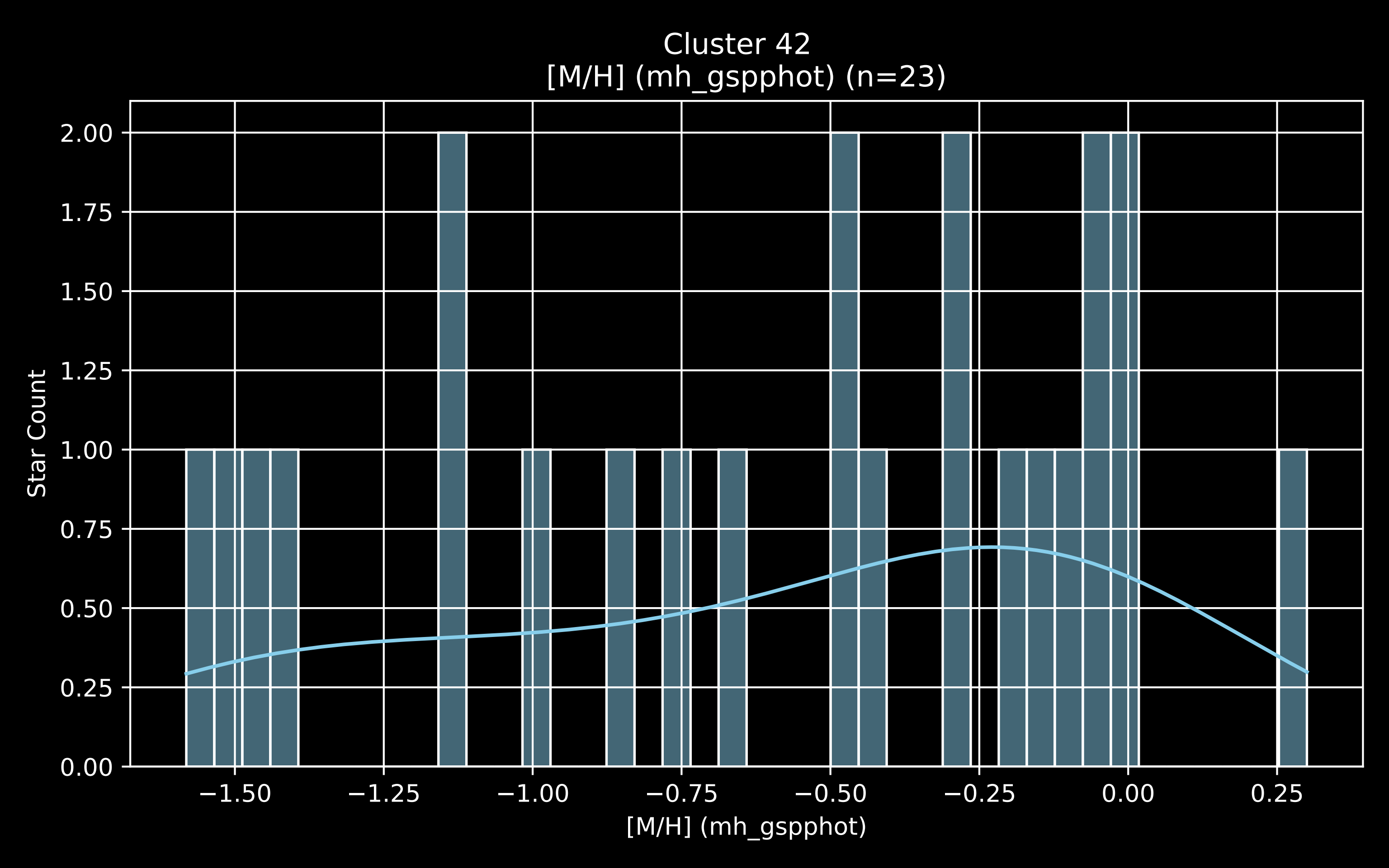
LAMOST DR11 Field: [GAC_063022N281243_M1] Age-Metallicity Relation (n= 581)
Stellar Age ($\log_{10}$(Age/yr))
Metallicity ([M/H]) (gspphot)
log(g) (Surface Gravity)
10.00
9.75
9.50
9.25
9.00
8.75
8.50
8.25
−4
−3
−2
−1
0
4.0
4.2
4.4
4.6
4.8
5.0
Median Cluster: $\log(Age) = 9.85, [Fe/H] = -0.43$

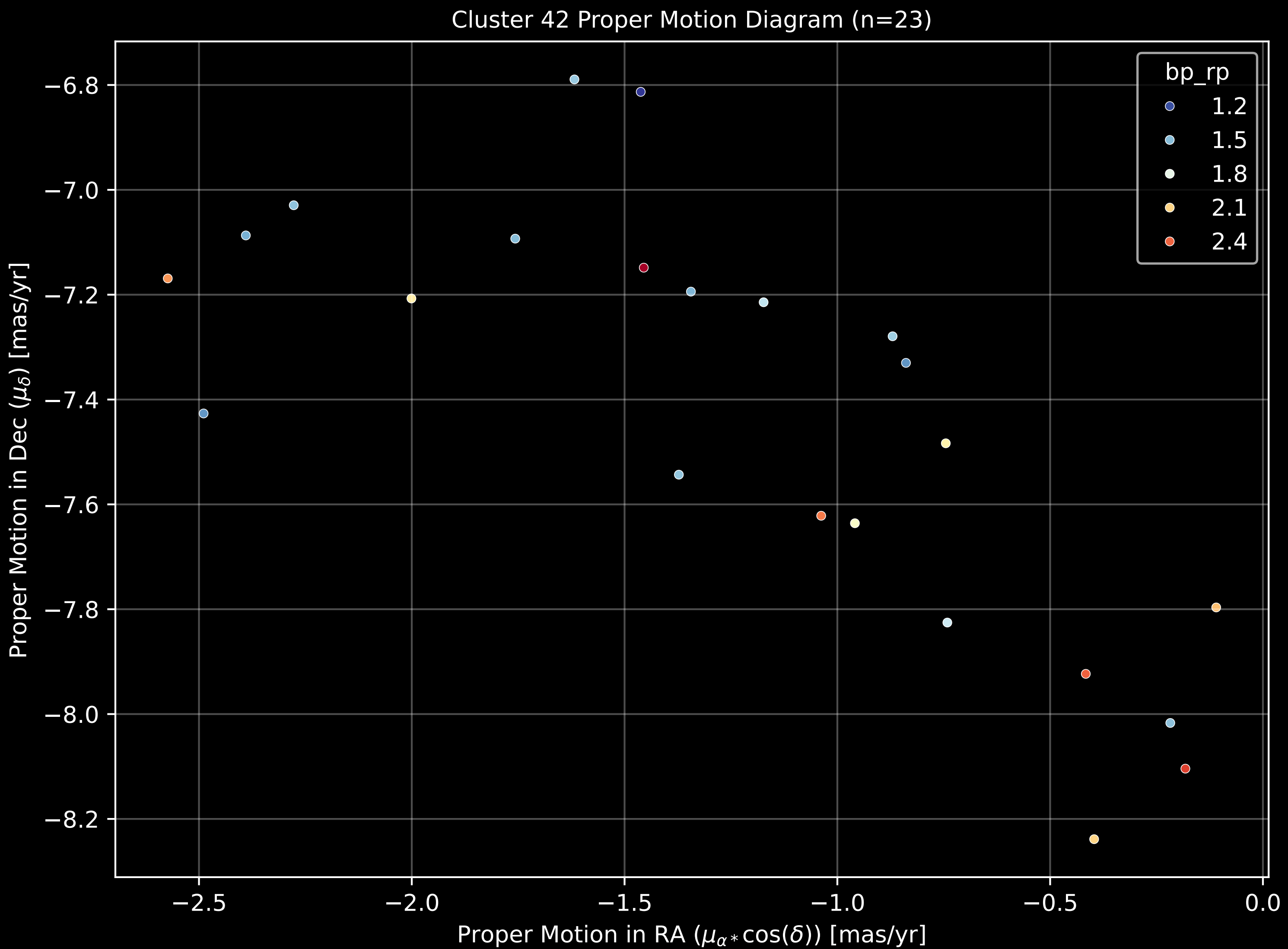

LAMOST DR11 Field [GAC_063022N281243_M1]
Proper Motion Vectors with HDBSCAN
(n=2023)
HDBSCAN Clusters
Stars
Cluster 42 (n=23)
Cluster 54 (n=20)
Cluster 96 (n=23)
Cluster 102 (n=23)
Cluster 105 (n=18)
Noise (n=1016)
Dec (deg)
RA (deg)
26
27
28
29
30
95
96
97
98
99
100



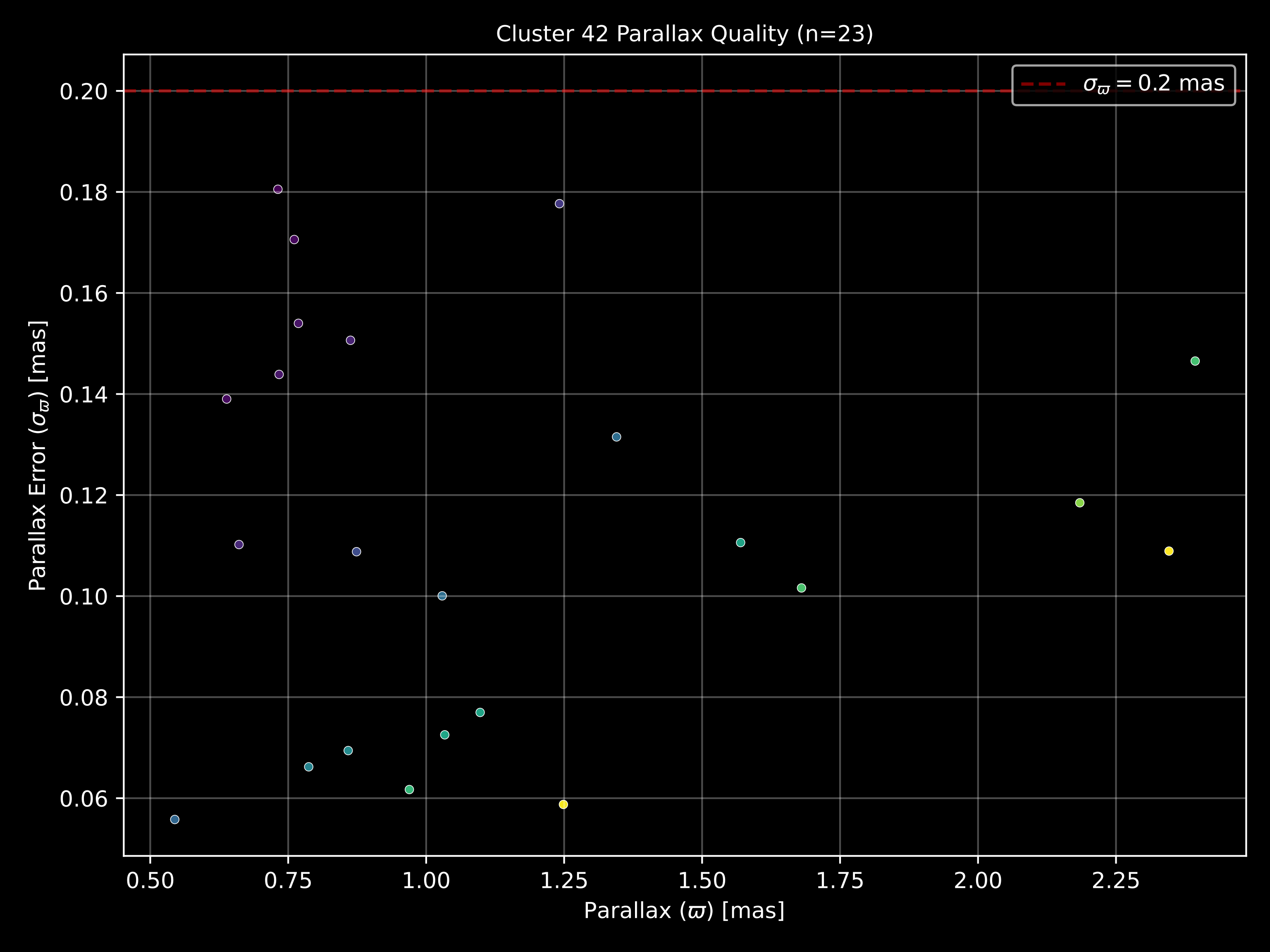
LAMOST DR11 Cluster 42 Proper Motion Vectors
(n=23)
Dec (deg)
RA (deg)
30.0
29.5
29.0
28.5
28.0
27.5
27.0
26.5
96
97
98
99
100

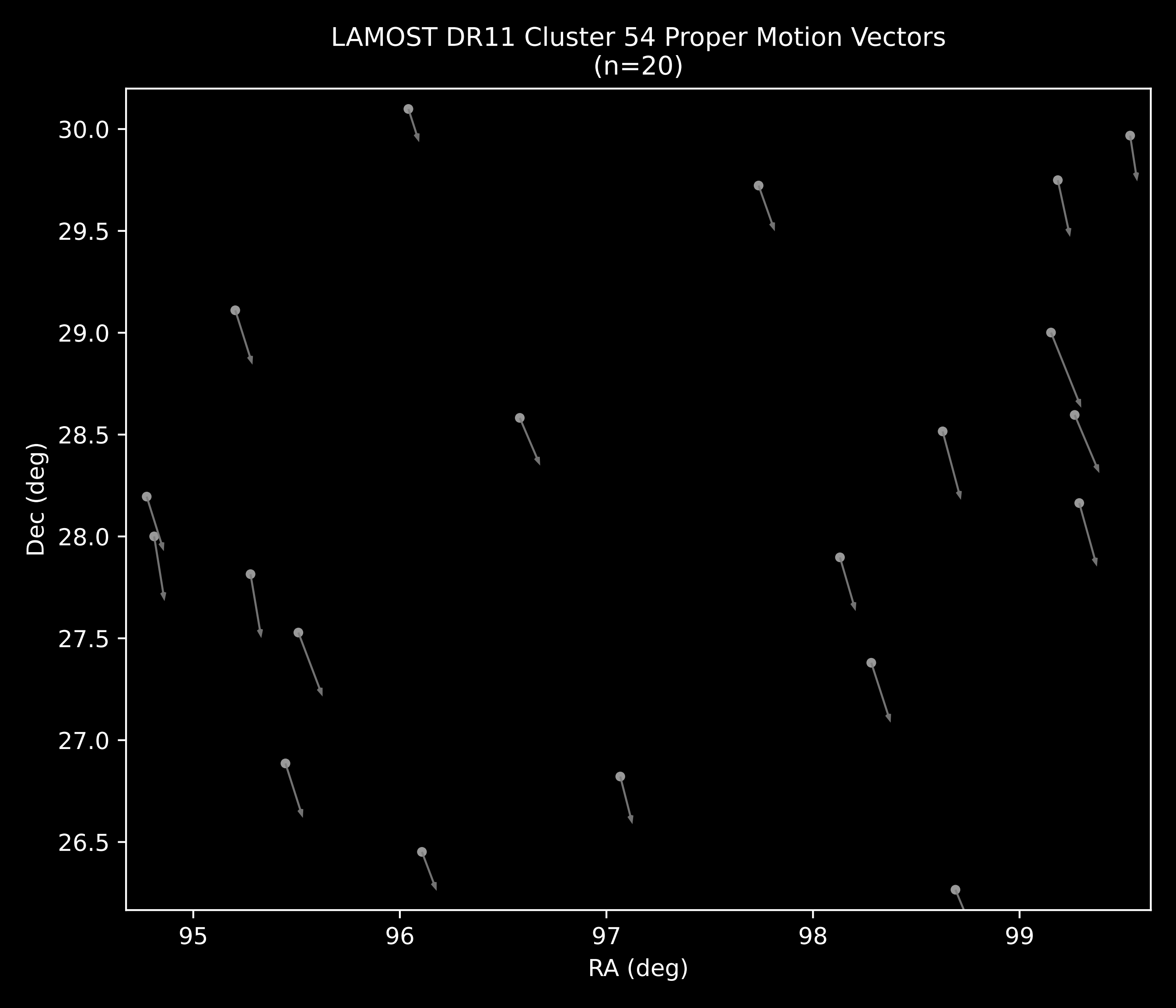

Cluster 42
Count plot of Star Type (n=23)
Count
20
15
10
5
0
main_sequence
Star Type


Cluster 42
[M/H] (mh_gspphot) (n=23)
Star Count
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
[M/H] (mh_gspphot)

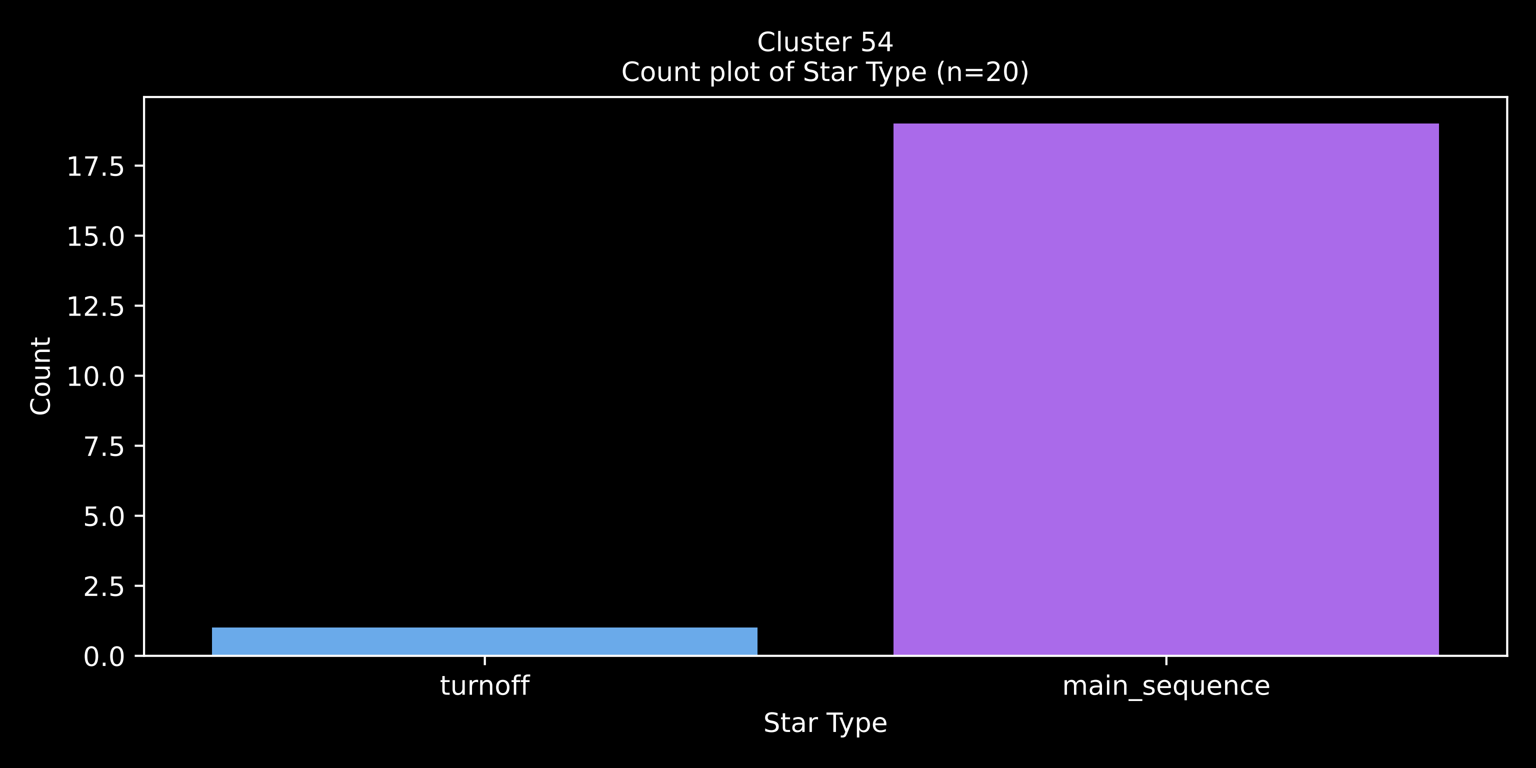


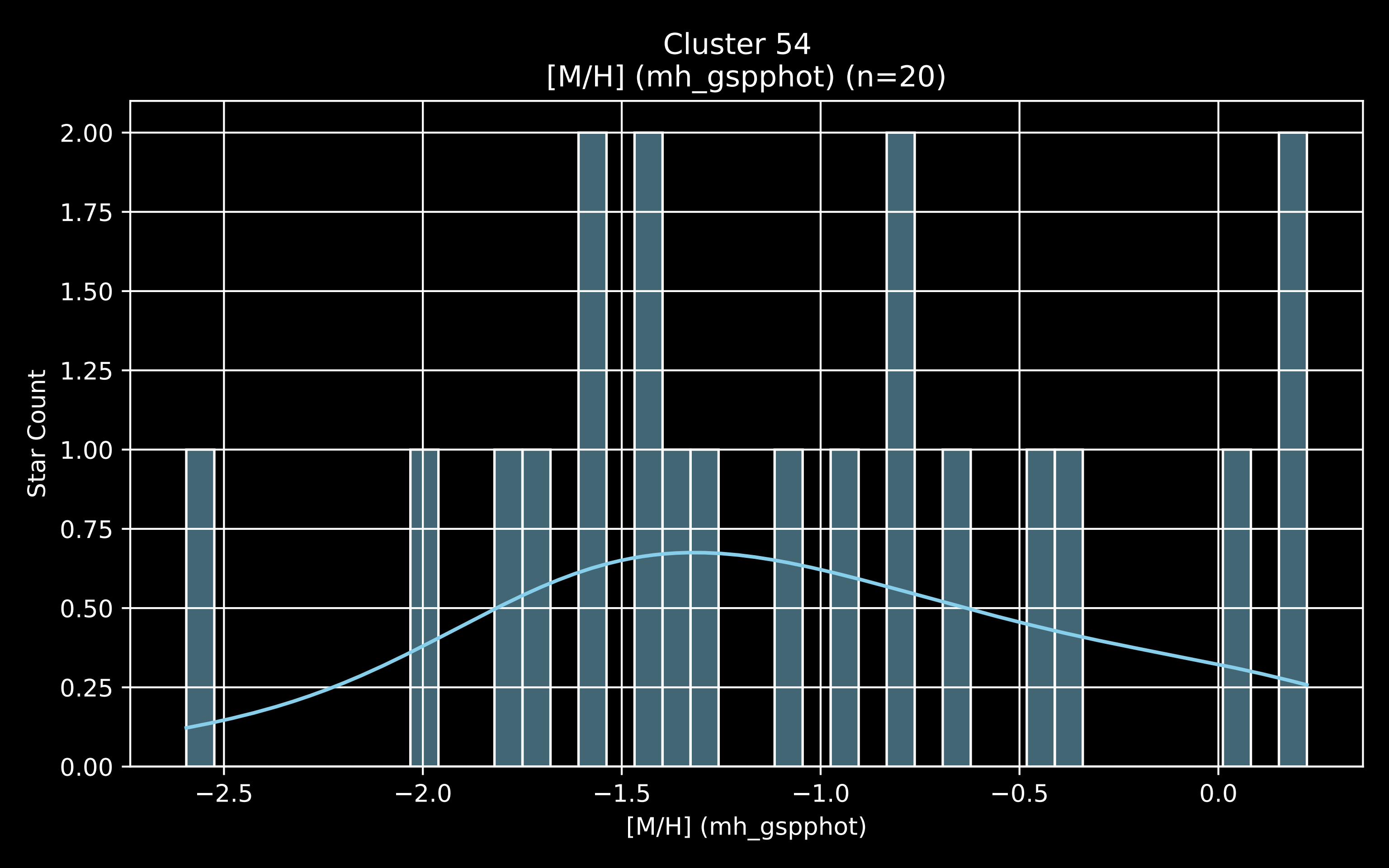
Cluster 42 Proper Motion Diagram (n=23)
bp_rp
1.2
1.5
1.8
2.1
2.4
−6.8
−7.0
−7.2
−7.4
−7.6
−7.8
−8.0
−8.2
Proper Motion in Dec (μ_δ) [mas/yr]
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

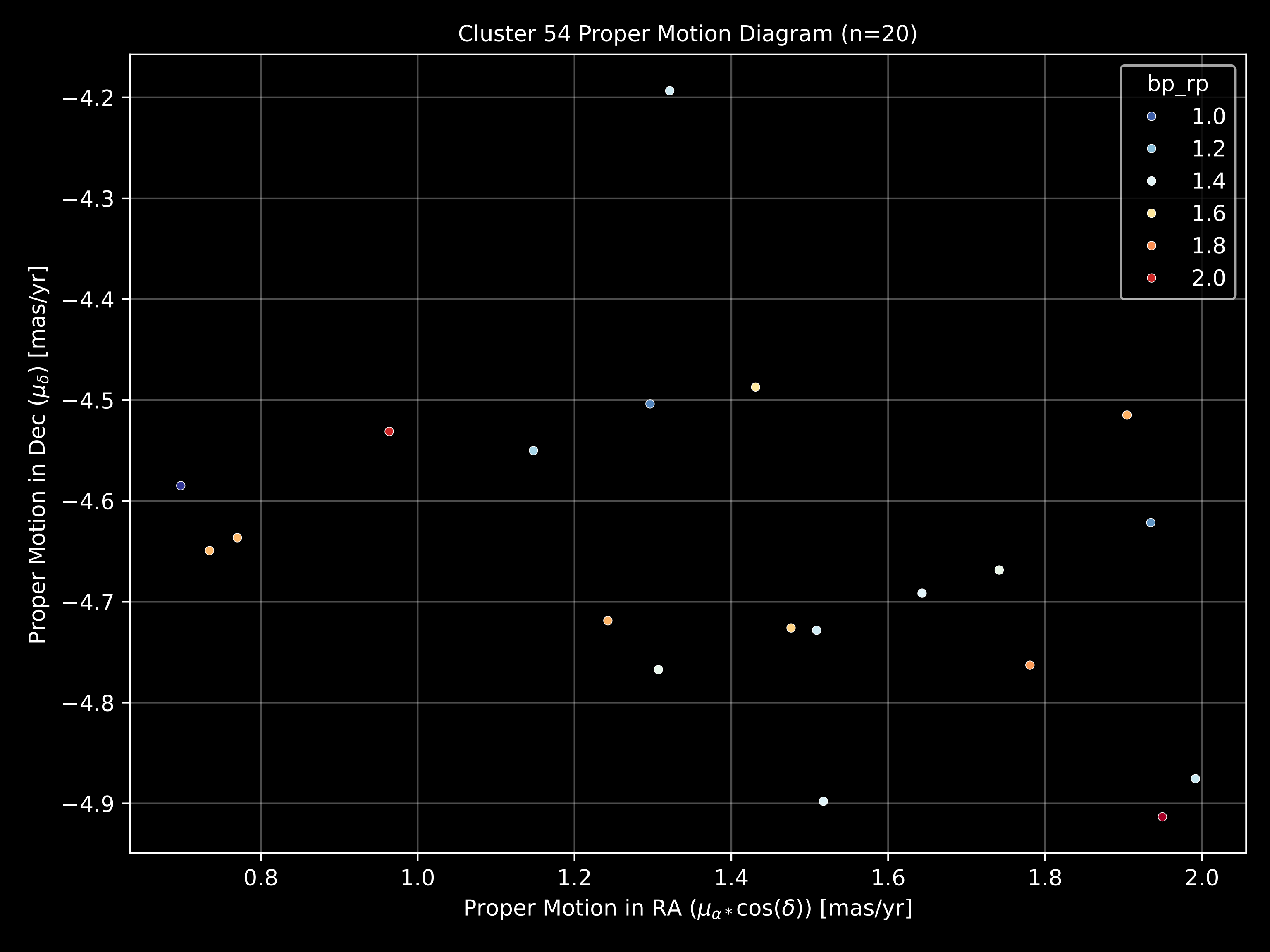

Cluster 42 Parallax Quality (n=23)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.06
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.50
0.75
1.00
1.25
1.50
1.75
2.00
2.25



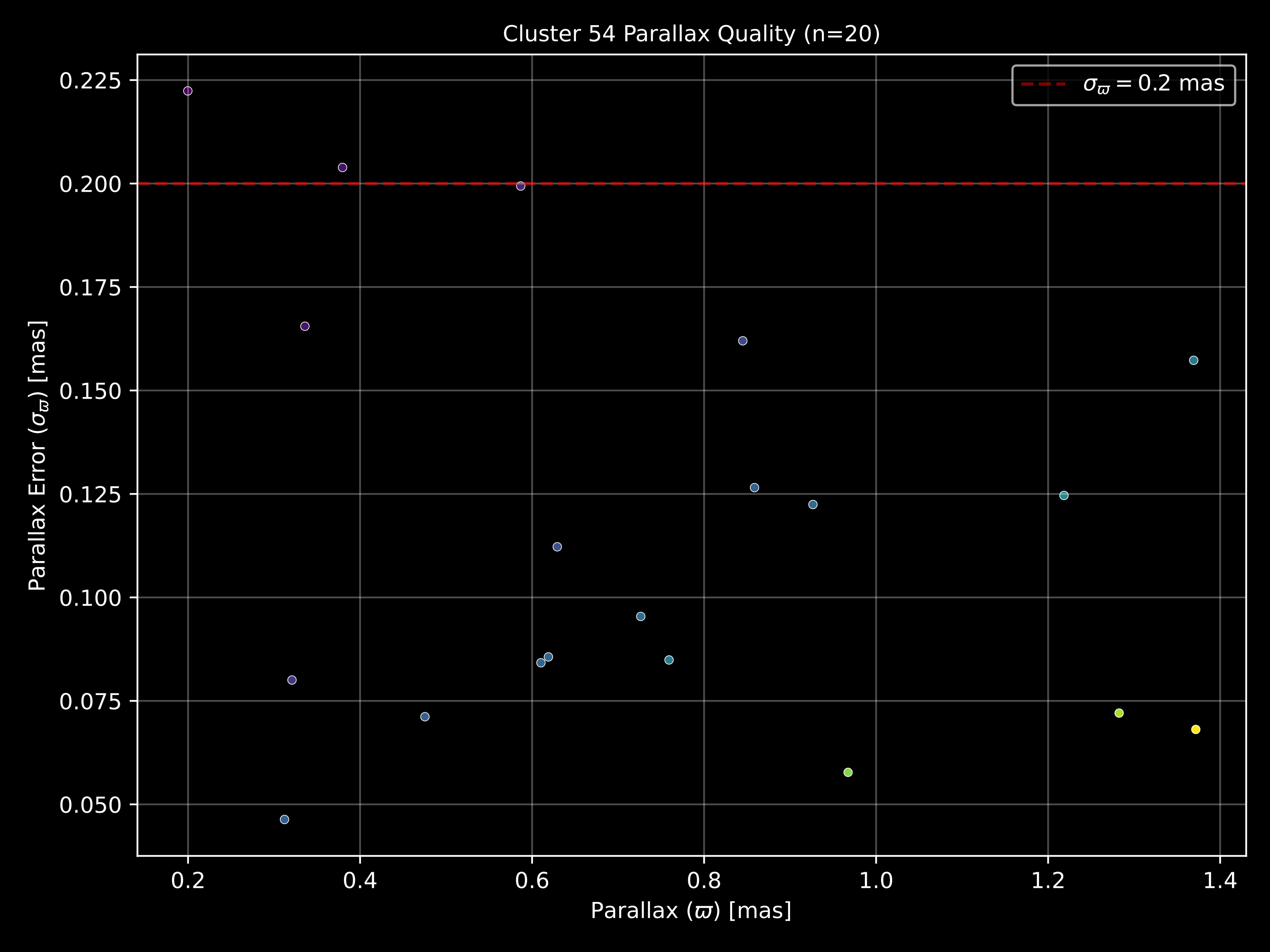
LAMOST DR11 Cluster 54 Proper Motion Vectors
(n=20)
Dec (deg)
RA (deg)
30.0
29.5
29.0
28.5
28.0
27.5
27.0
26.5
95
96
97
98
99


Cluster 54
Count plot of Star Type (n=20)
Count
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
turnoff
main_sequence
Star Type

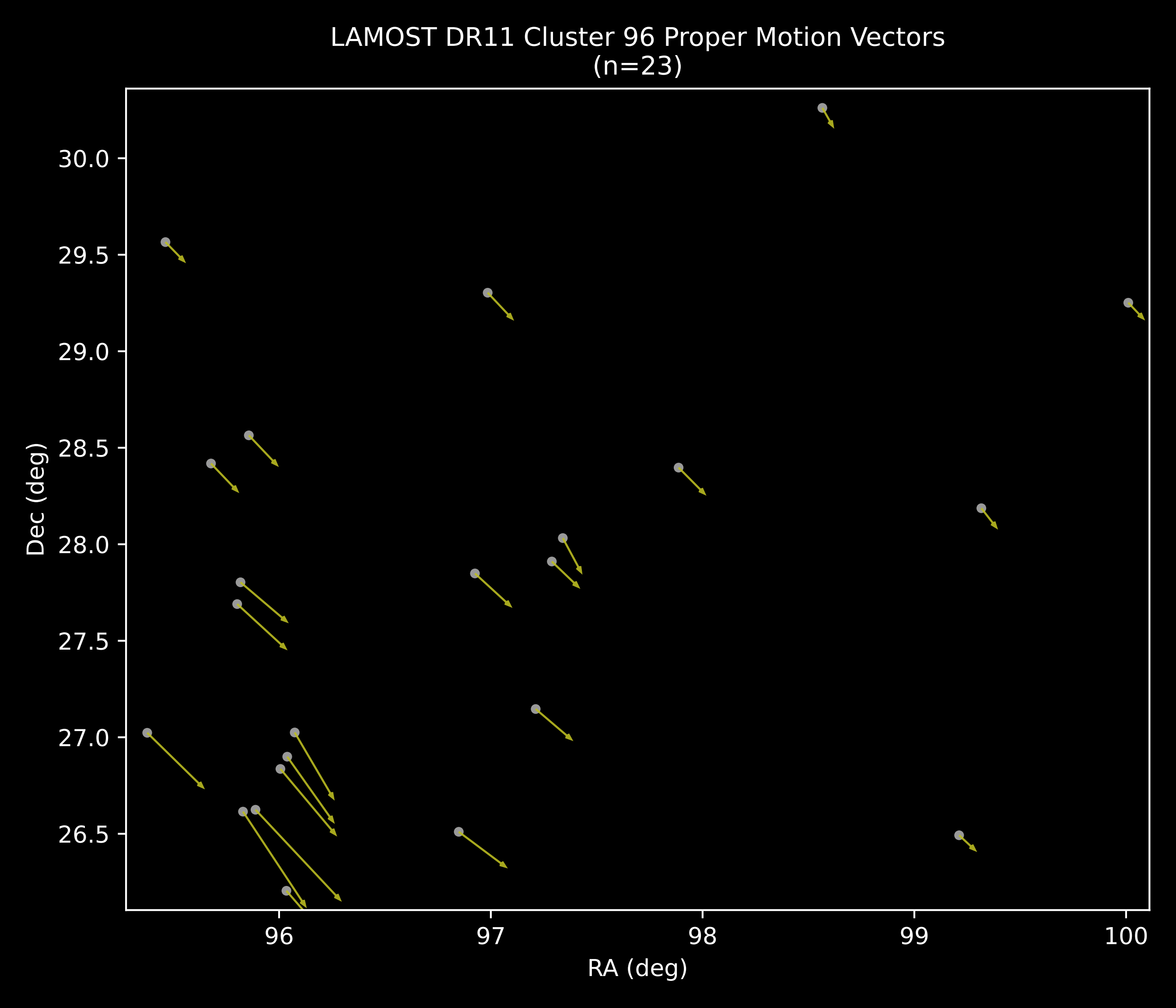

Cluster 54
[M/H] (mh_gspphot) (n=20)
Star Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
[M/H] (mh_gspphot)

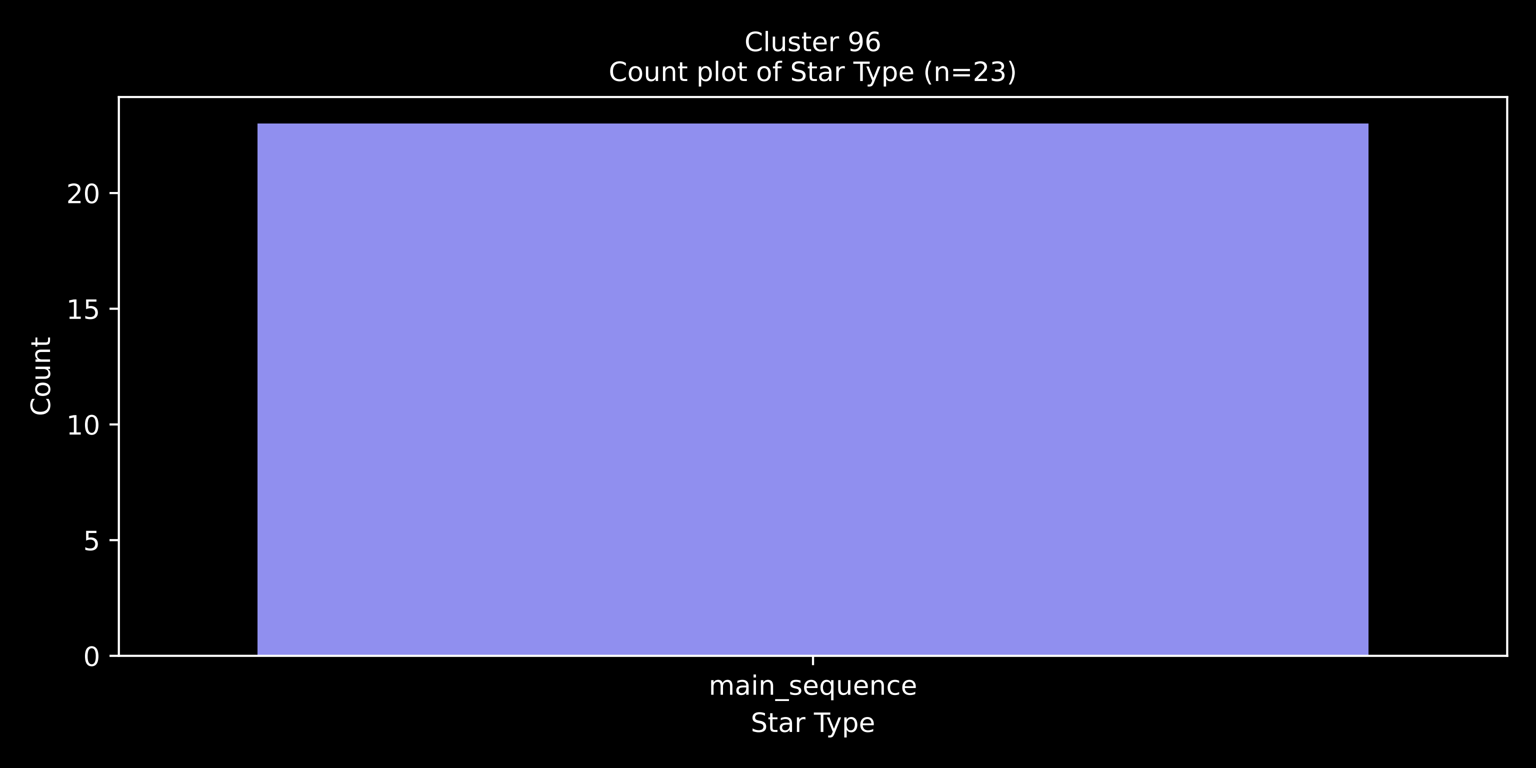


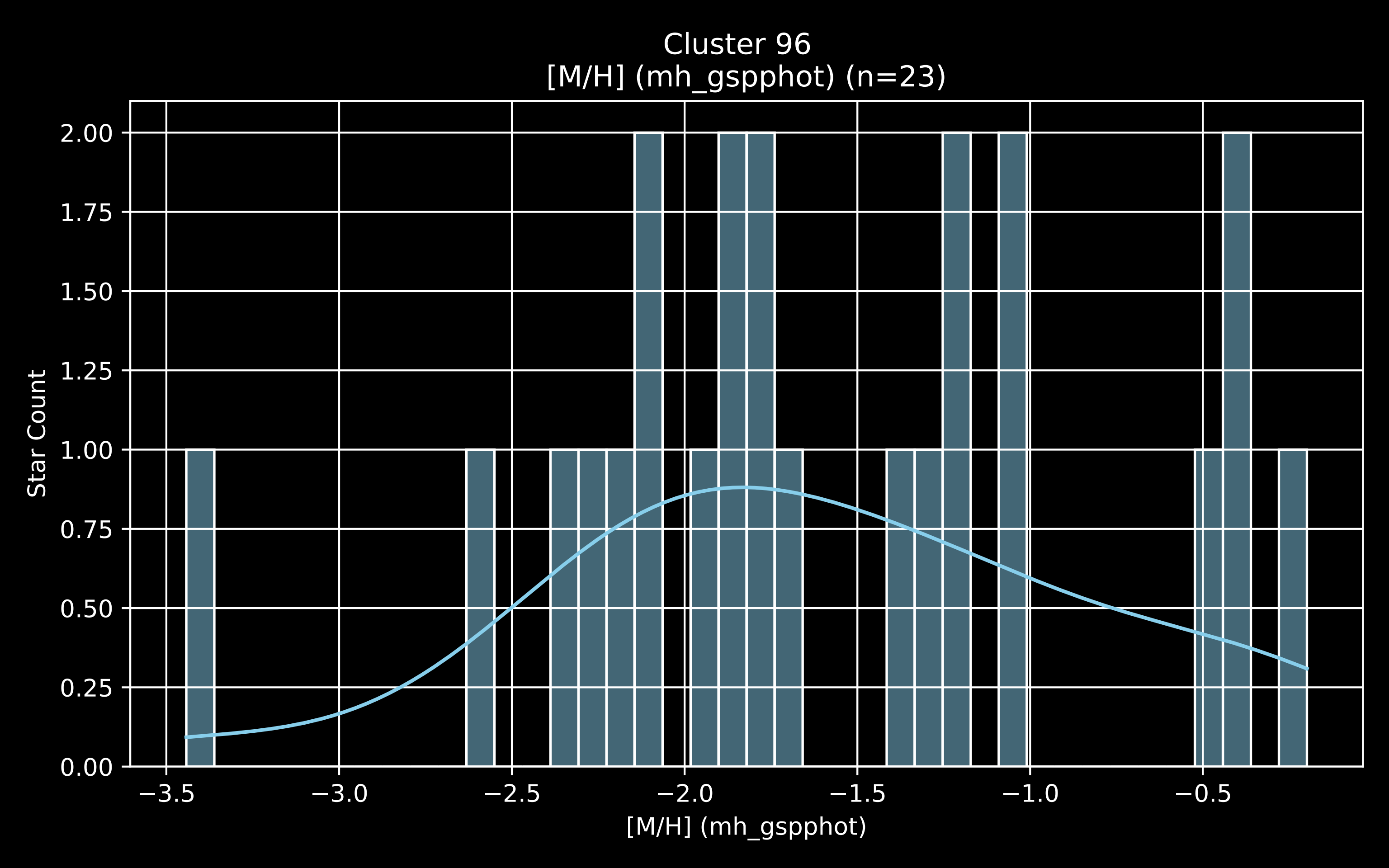
Cluster 54 Proper Motion Diagram (n=20)
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
bp_rp
1.0
1.2
1.4
1.6
1.8
2.0
−4.2
−4.3
−4.4
−4.5
−4.6
−4.7
−4.8
−4.9
0.8
1.0
1.2
1.4
1.6
1.8
2.0



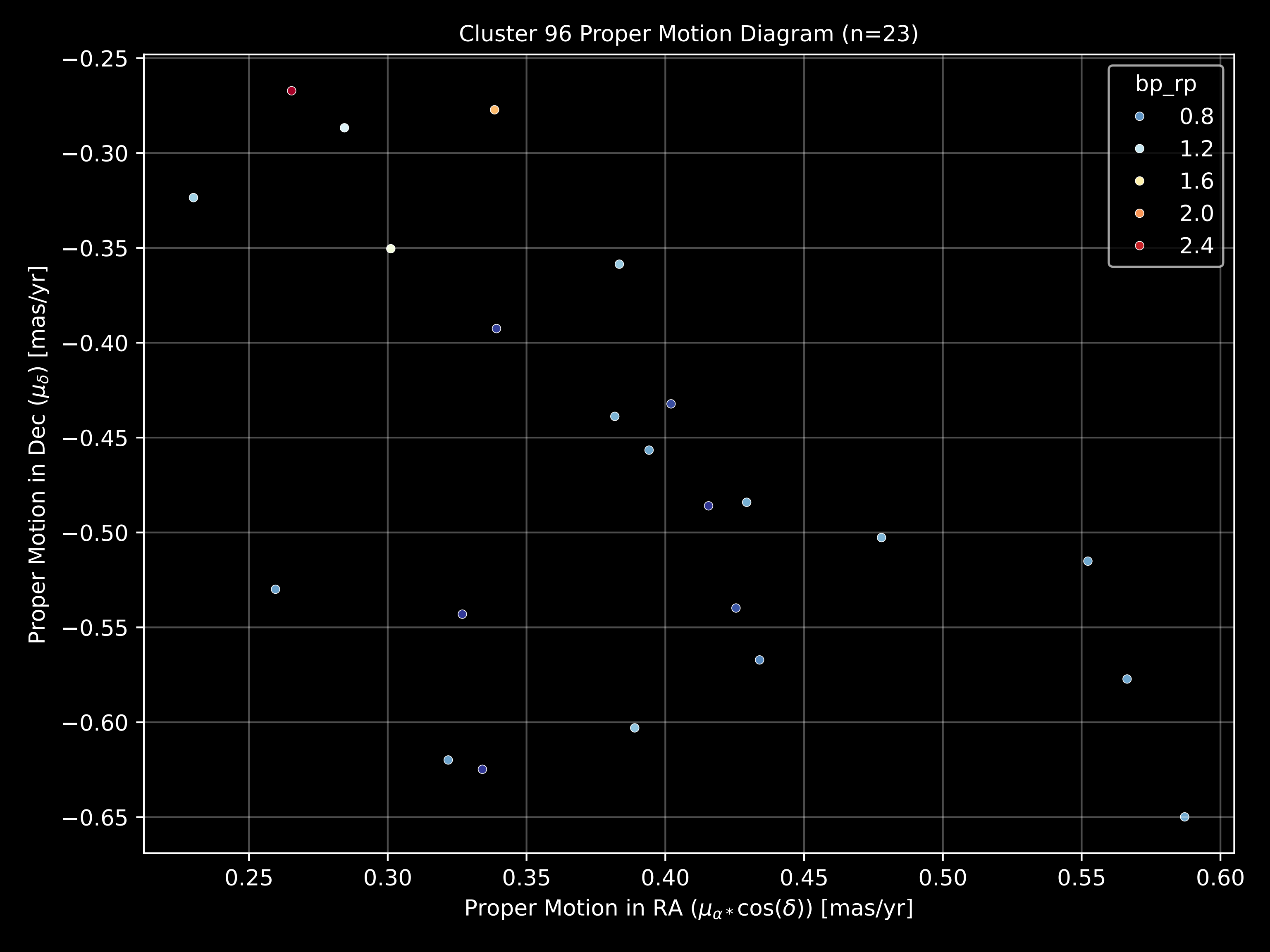
Cluster 54 Parallax Quality (n=20)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.2
0.4
0.6
0.8
1.0
1.2
1.4


LAMOST DR11 Cluster 96 Proper Motion Vectors
(n=23)
Dec (deg)
RA (deg)
30.0
29.5
29.0
28.5
28.0
27.5
27.0
26.5
96
97
98
99
100

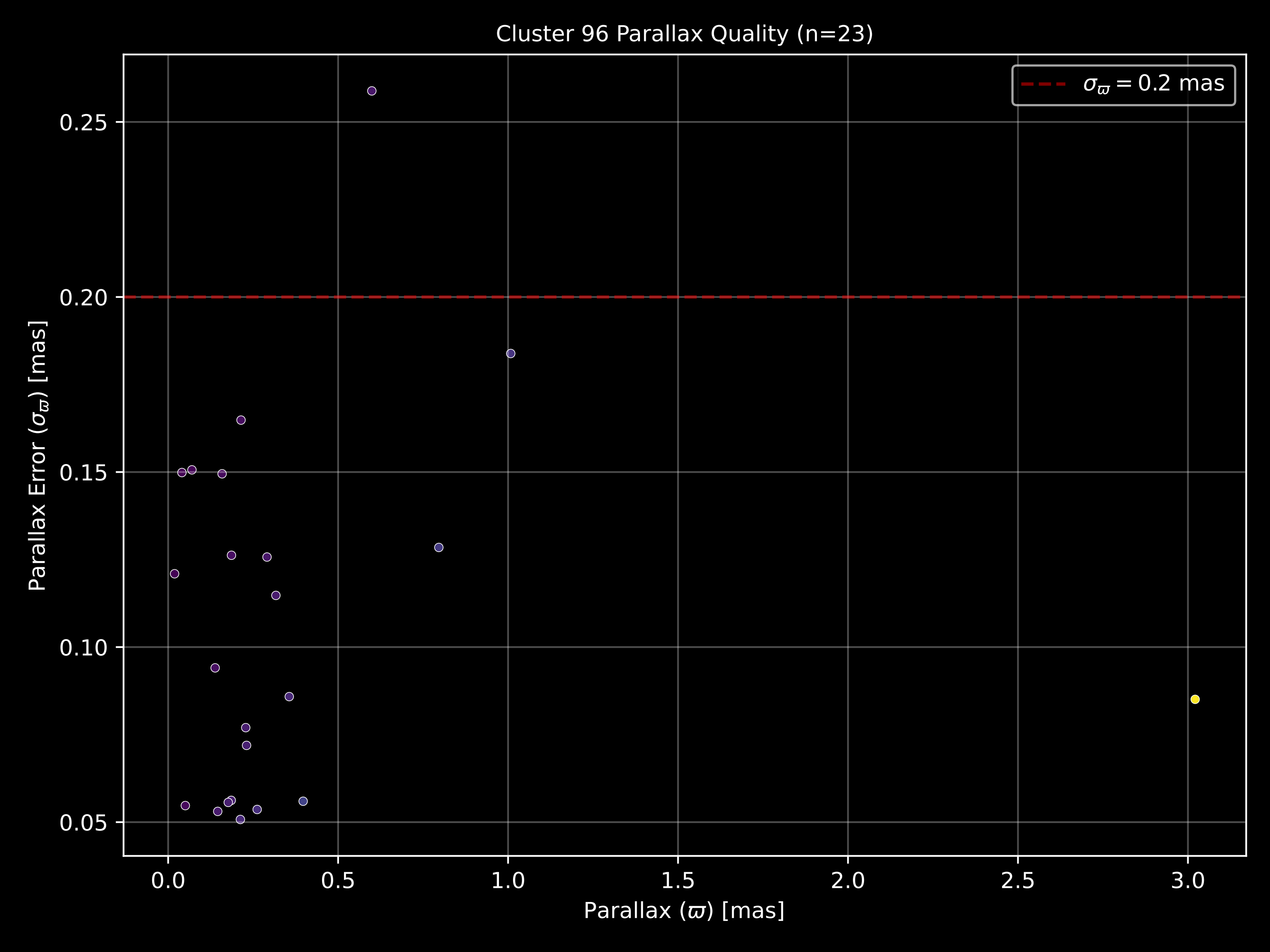

Cluster 96
Count plot of Star Type (n=23)
Count
20
15
10
5
0
main_sequence
Star Type



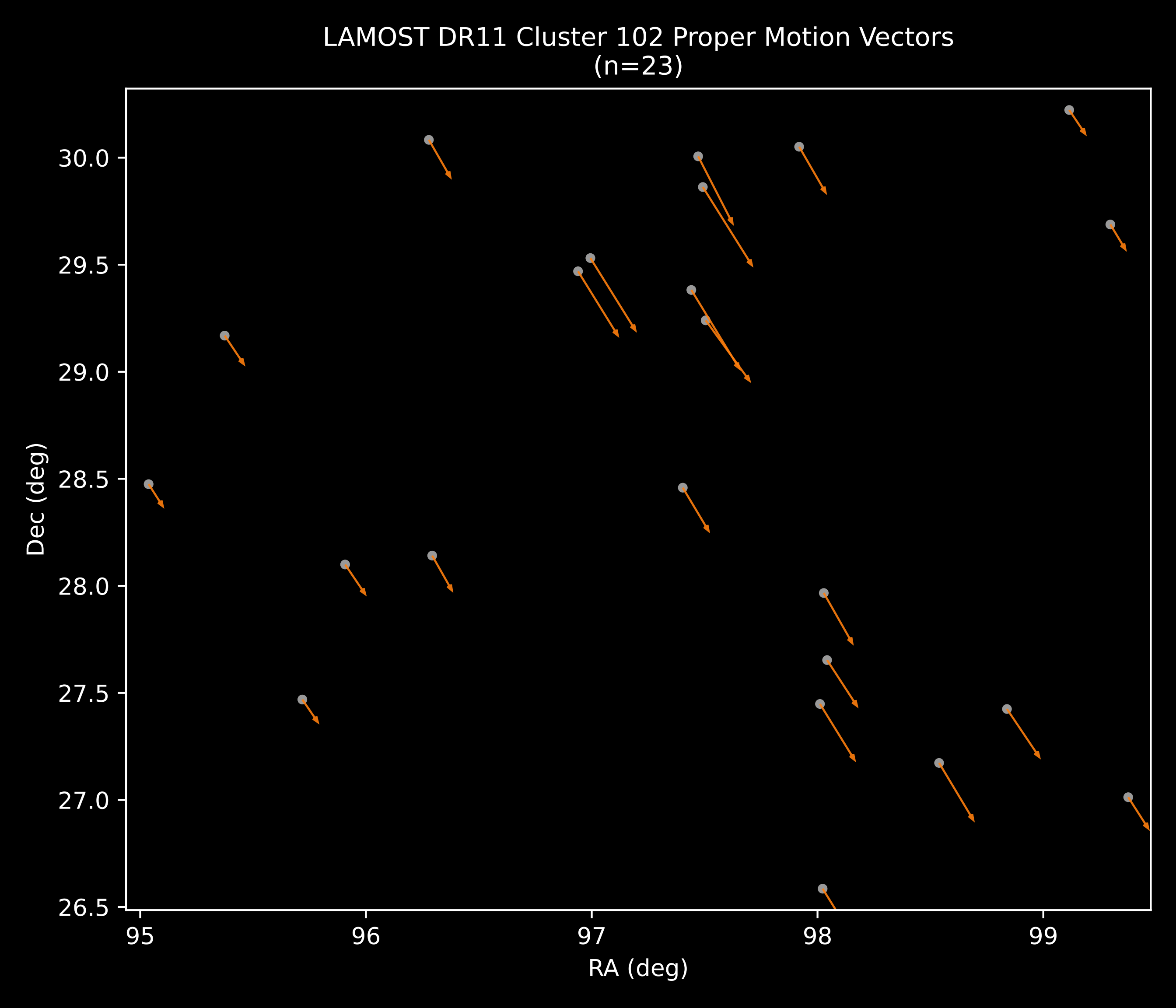
Cluster 96
[M/H] (mh_gspphot) (n=23)
Star Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
[M/H] (mh_gspphot)

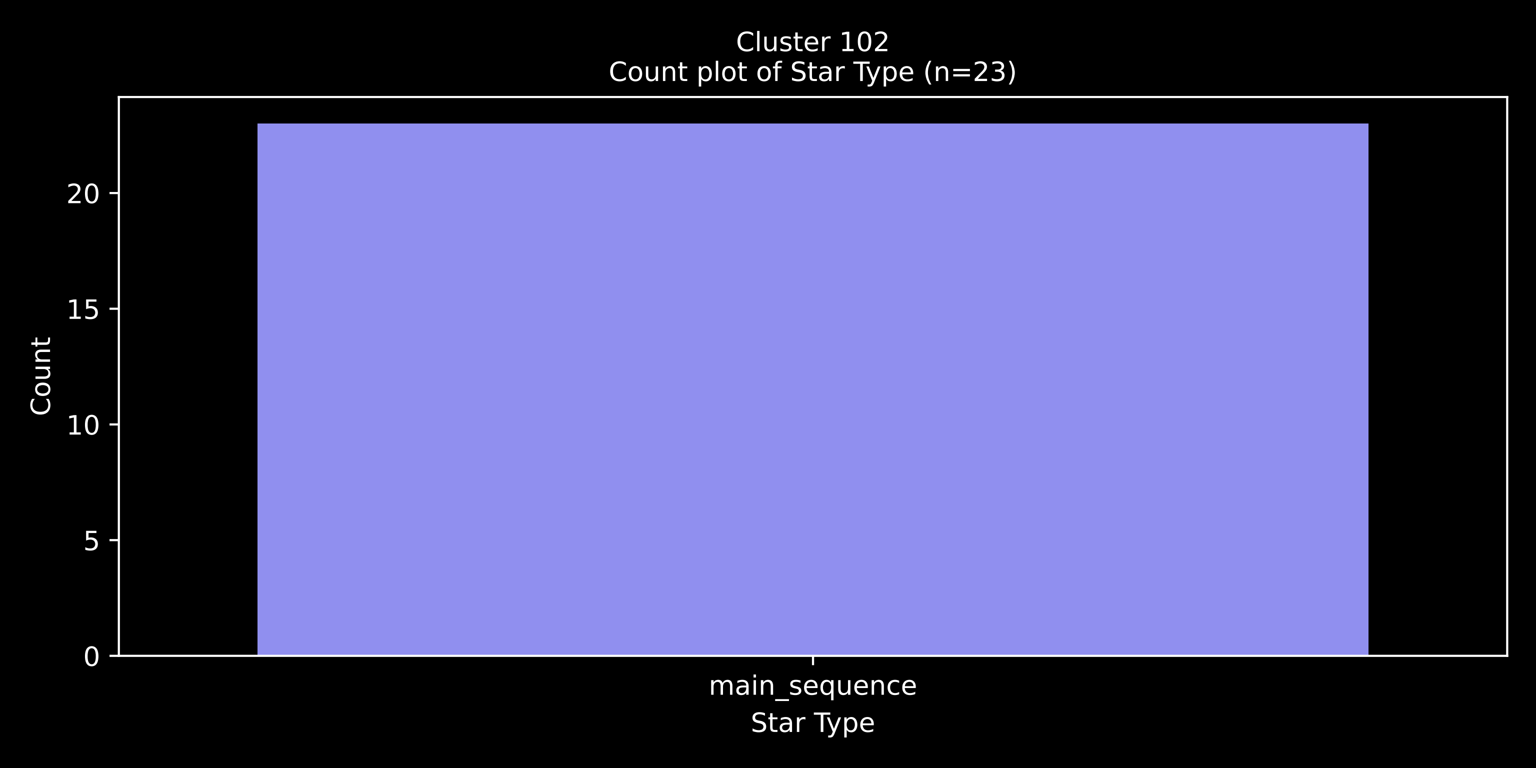


Cluster 96 Proper Motion Diagram (n=23)
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
bp_rp
0.8
1.2
1.6
2.0
2.4
−0.25
−0.30
−0.35
−0.40
−0.45
−0.50
−0.55
−0.60
−0.65
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60

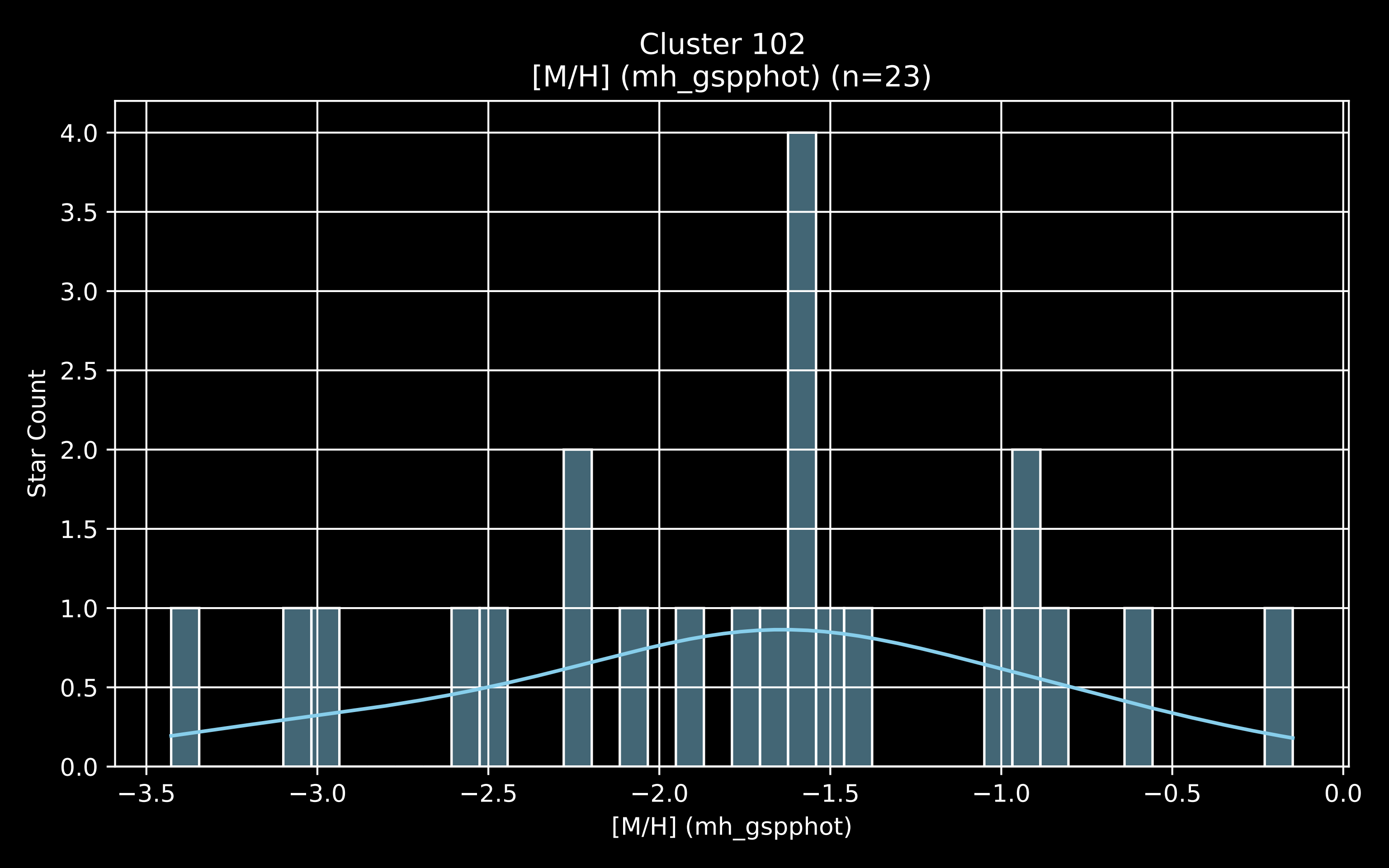


Cluster 96 Parallax Quality (n=23)
$\sigma_\varpi = 0.2$ mas
0.25
0.20
0.15
0.10
0.05
Parallax Error (σ_ϖ) [mas]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Parallax (ϖ) [mas]

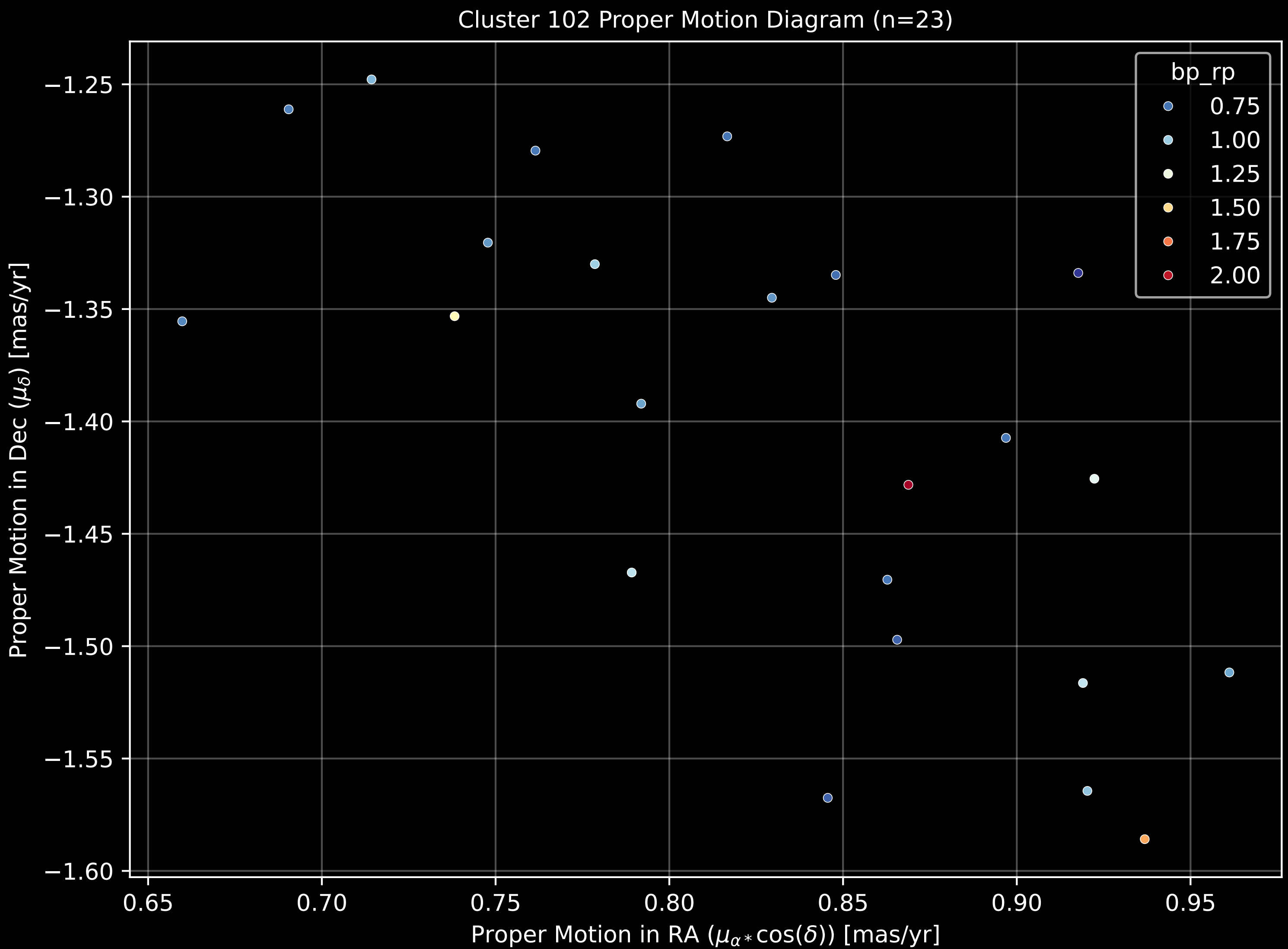

LAMOST DR11 Cluster 102 Proper Motion Vectors
(n=23)
Dec (deg)
RA (deg)
30.0
29.5
29.0
28.5
28.0
27.5
27.0
26.5
95
96
97
98
99



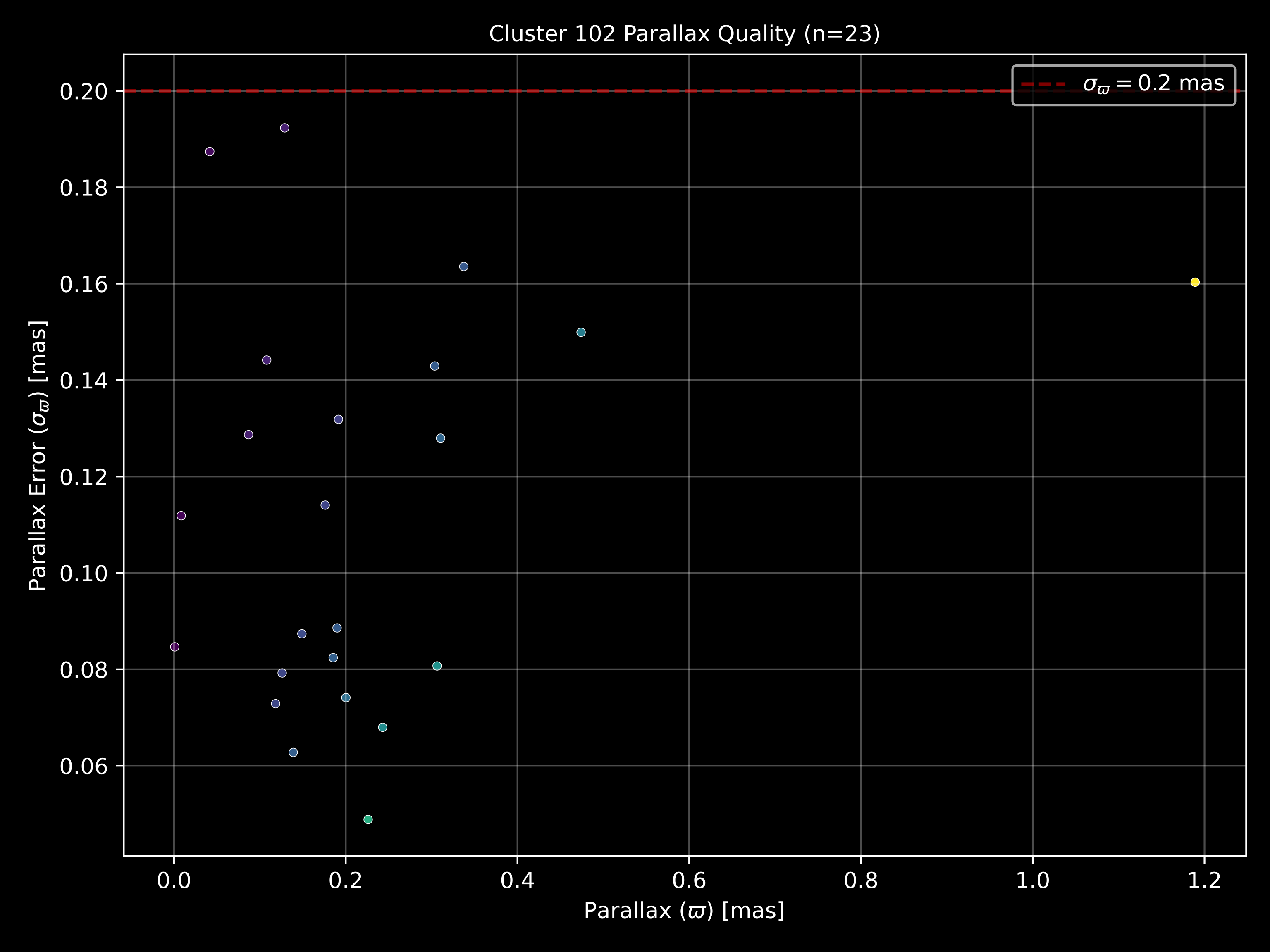
Cluster 102
Count plot of Star Type (n=23)
Count
20
15
10
5
0
main_sequence
Star Type



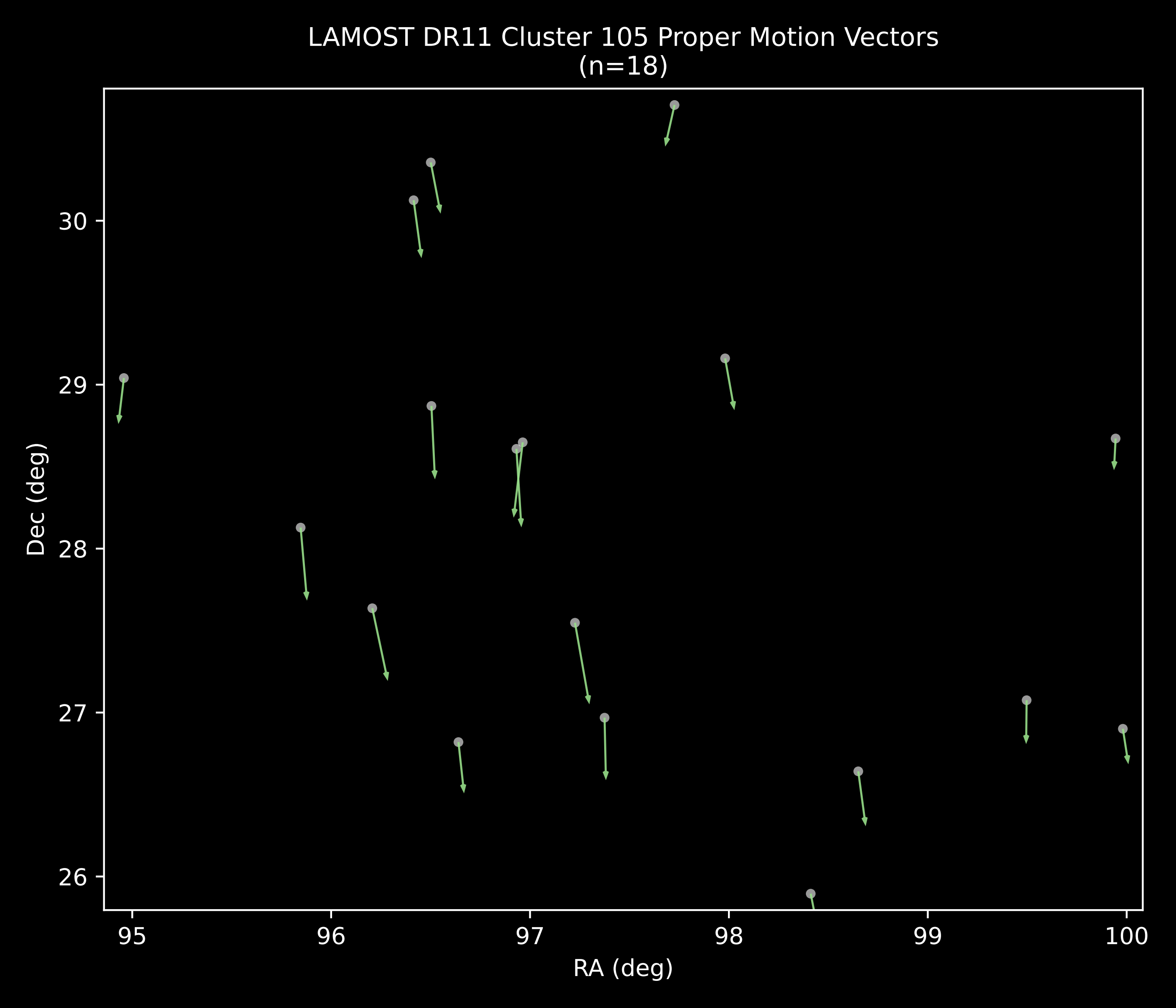
Cluster 102
[M/H] (mh_gspphot) (n=23)
Star Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
[M/H] (mh_gspphot)

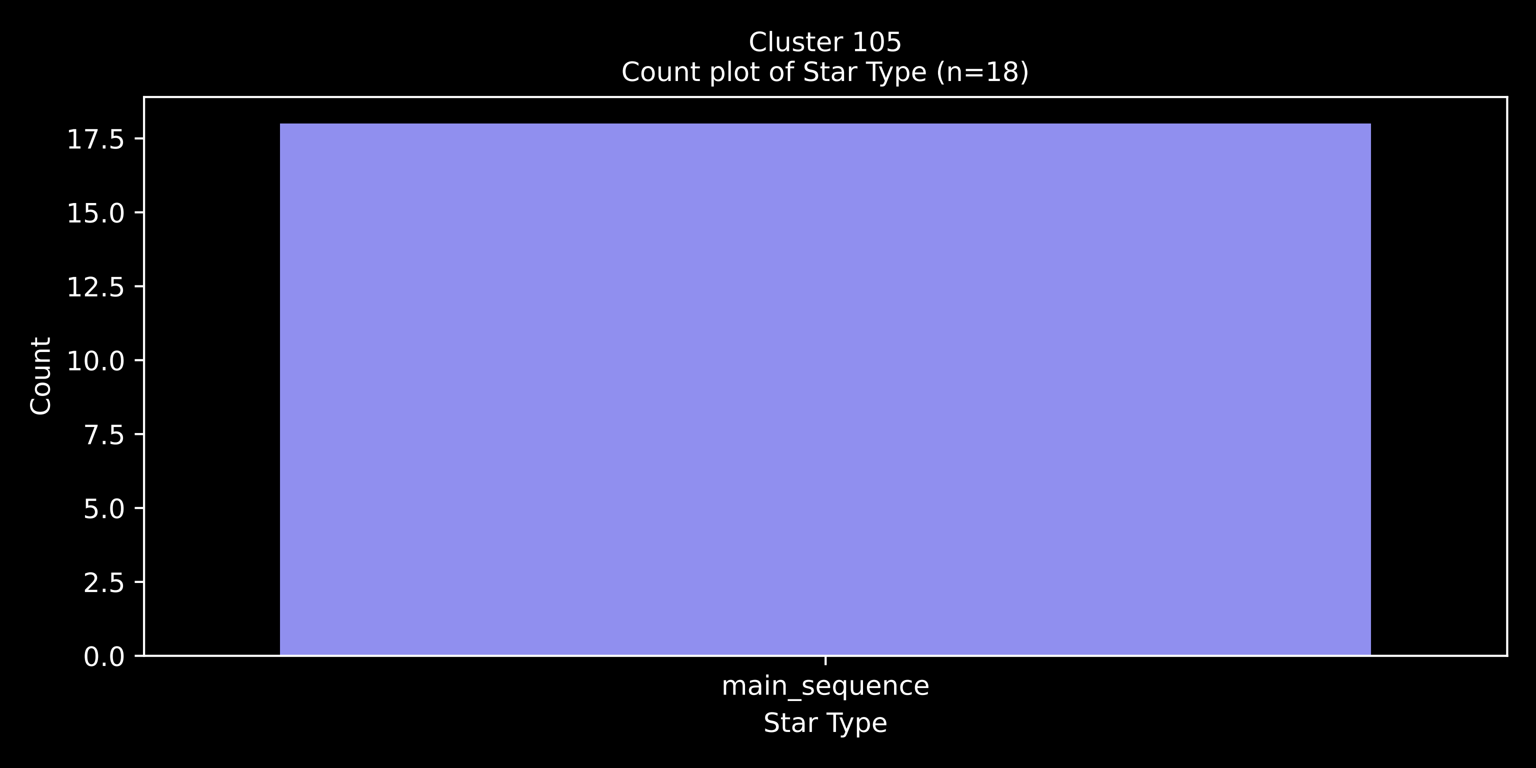


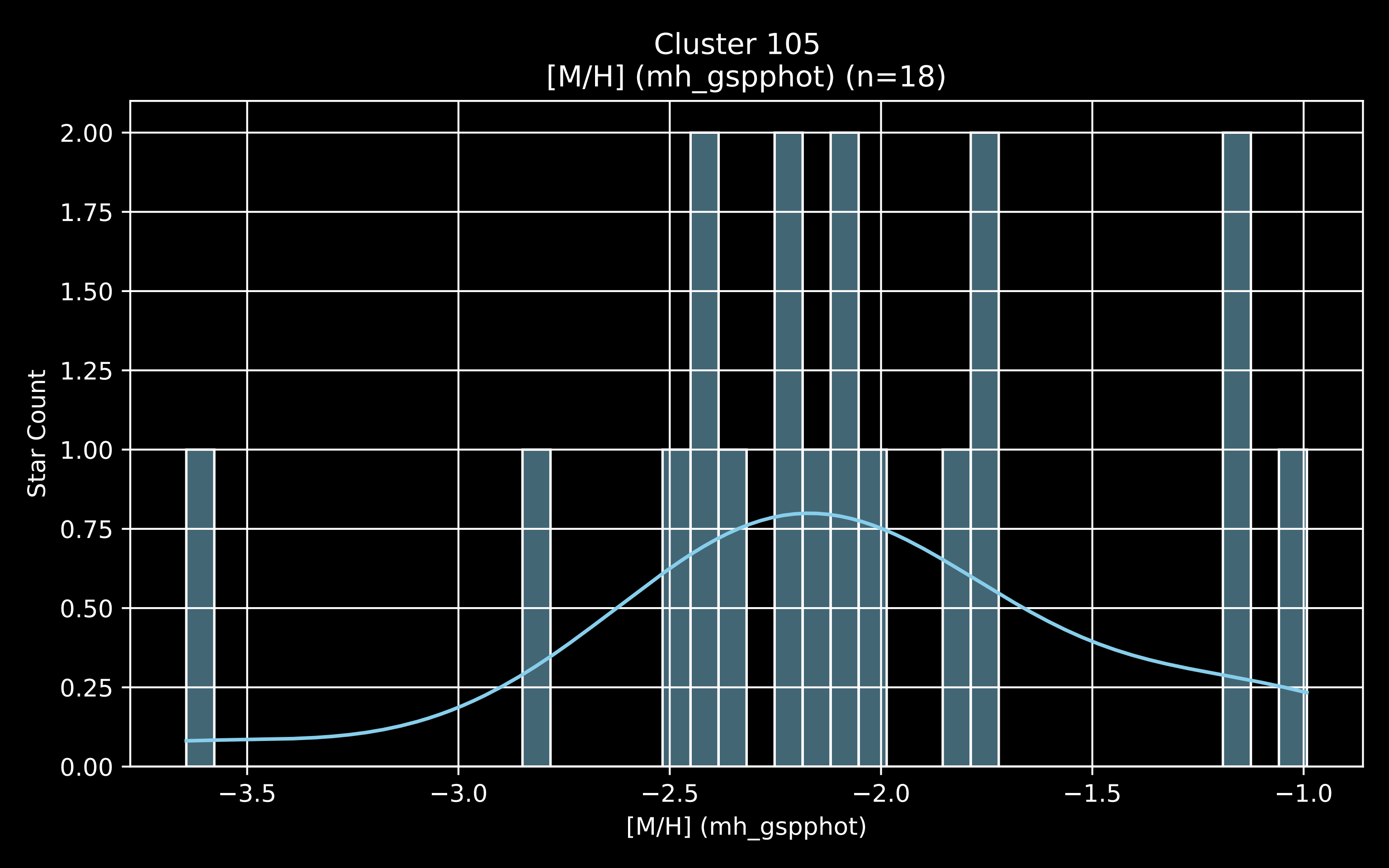
Cluster 102 Proper Motion Diagram (n=23)
bp_rp
0.75
1.00
1.25
1.50
1.75
2.00
−1.25
−1.30
−1.35
−1.40
−1.45
−1.50
−1.55
−1.60
Proper Motion in Dec (μ_δ) [mas/yr]
0.65
0.70
0.75
0.80
0.85
0.90
0.95
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]



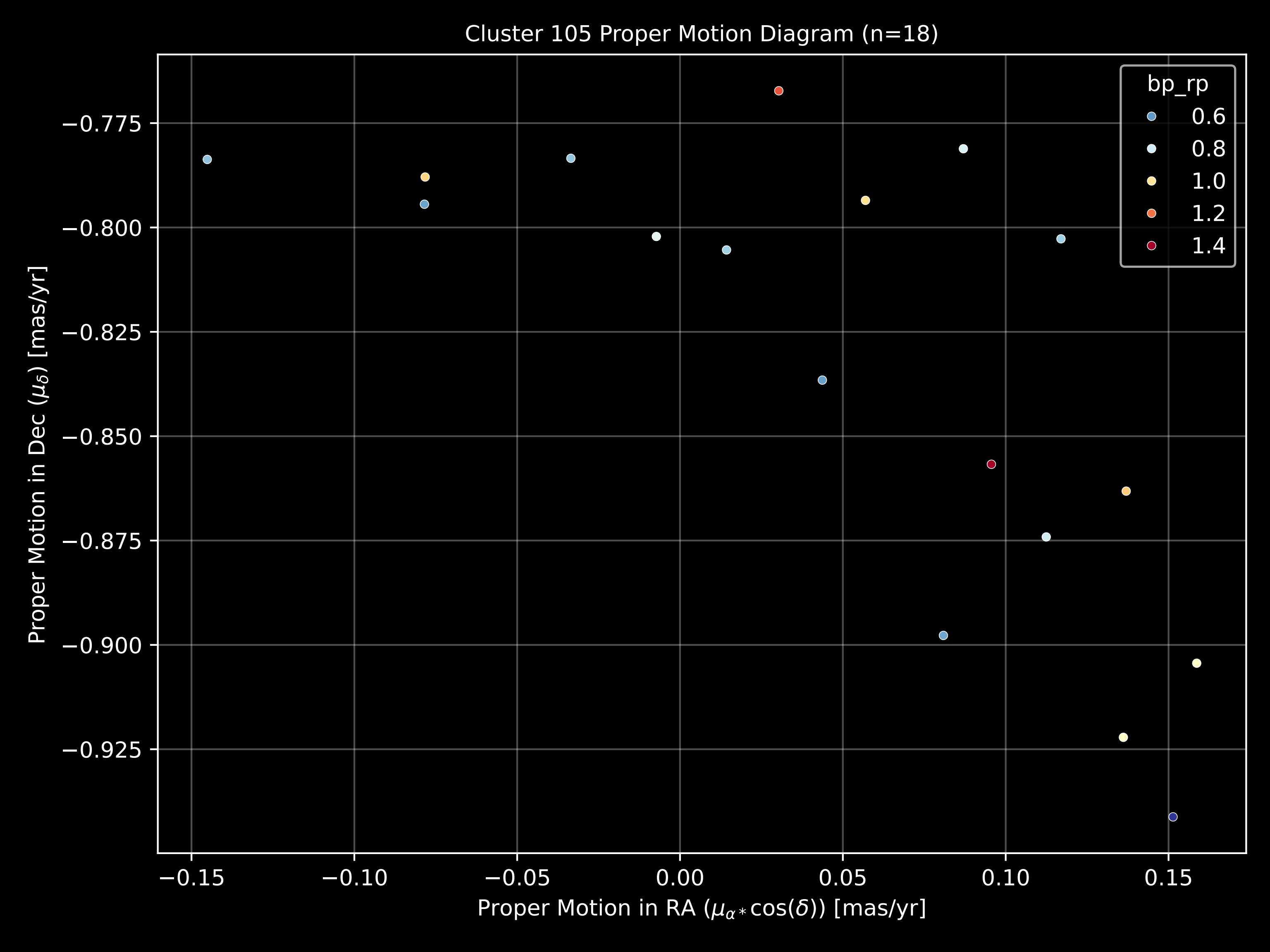
Cluster 102 Parallax Quality (n=23)
$\sigma_{\varpi} = 0.2$ mas
Parallax Error (σ_{ϖ}) [mas]
Parallax (ϖ) [mas]
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.0
0.2
0.4
0.6
0.8
1.0
1.2


LAMOST DR11 Cluster 105 Proper Motion Vectors
(n=18)
Dec (deg)
RA (deg)
30
29
28
27
26
95
96
97
98
99
100

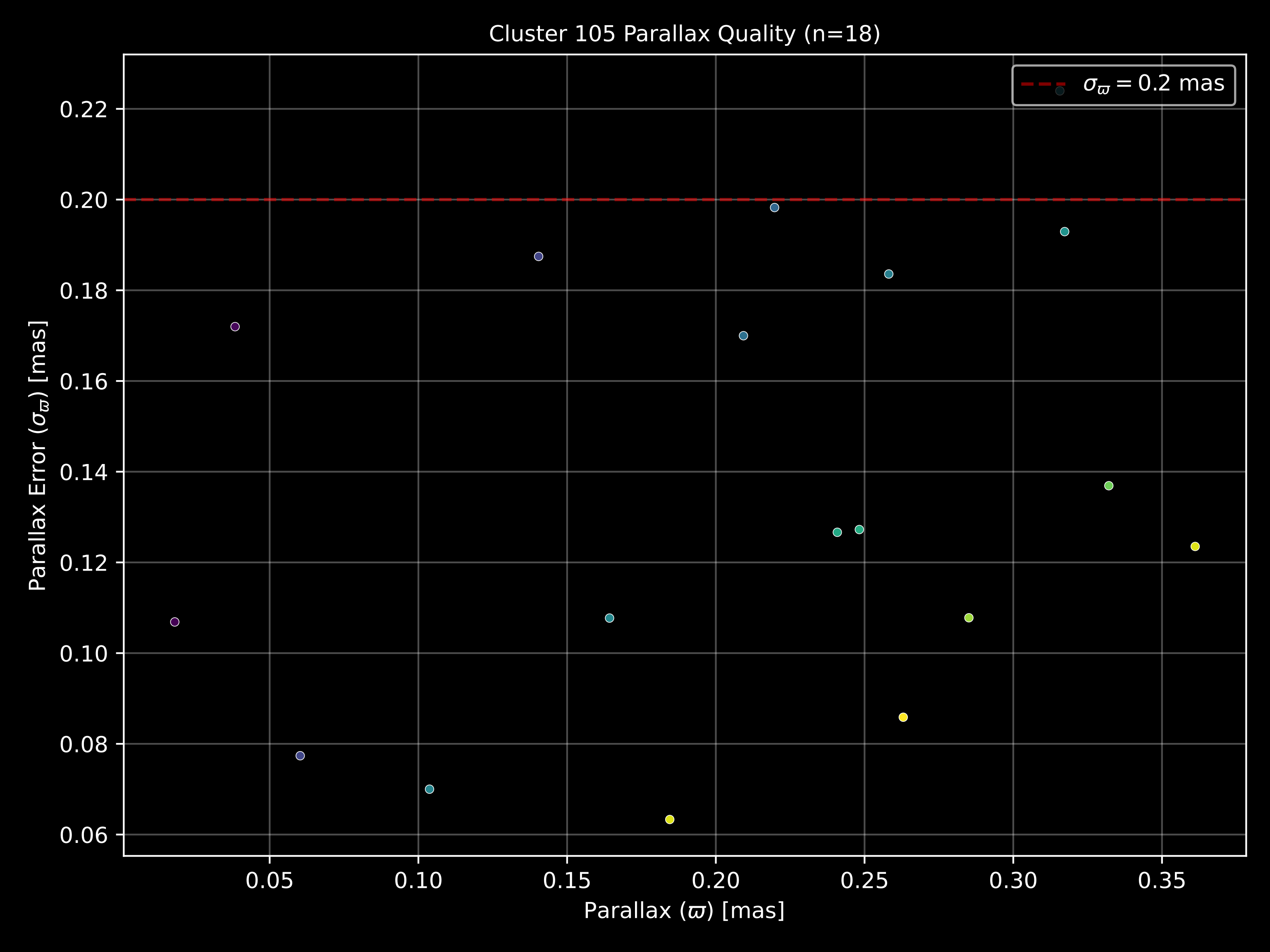

Cluster 105
Count plot of Star Type (n=18)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Count
main_sequence
Star Type

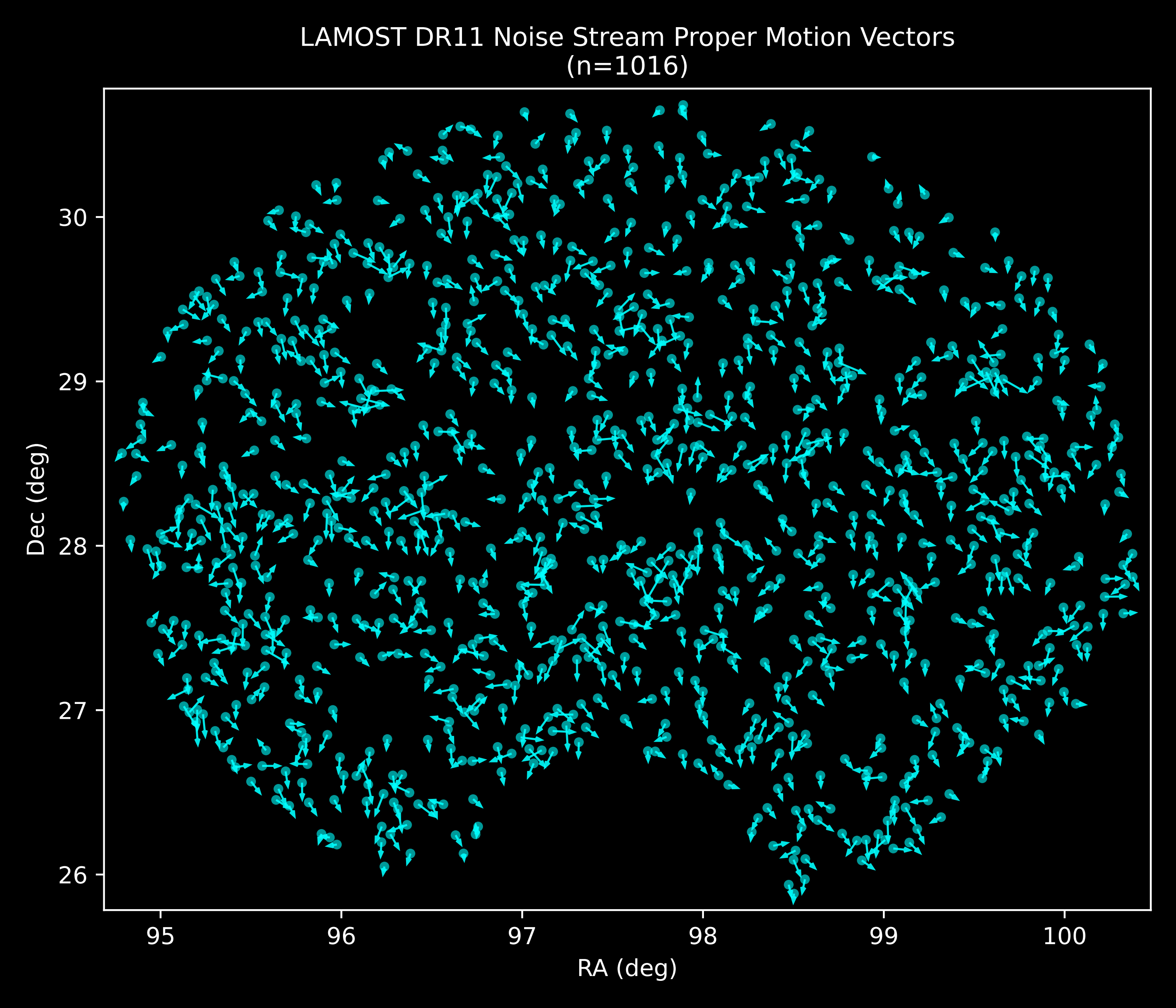


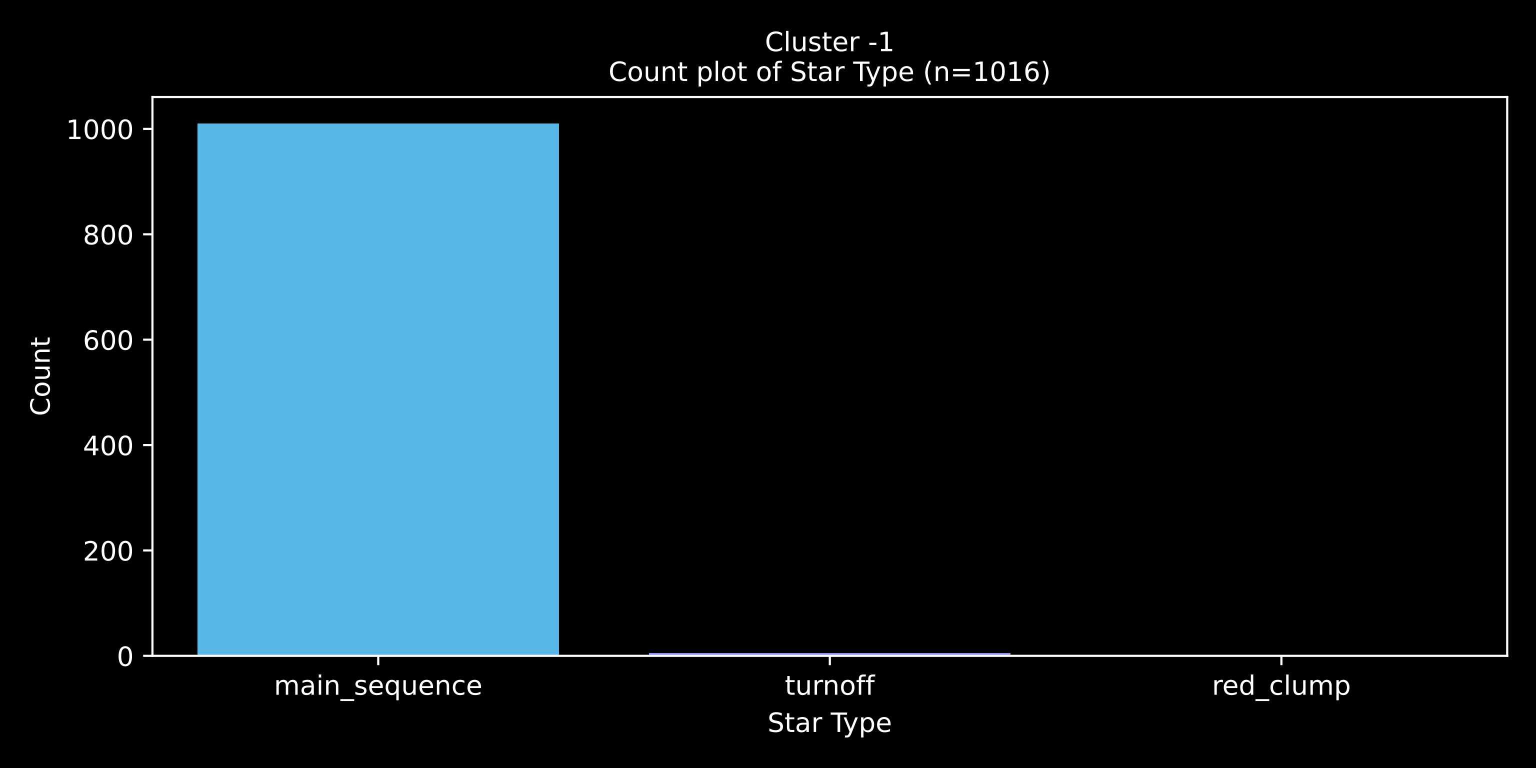
Cluster 105
[M/H] (mh_gspphot) (n=18)
Star Count
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
[M/H] (mh_gspphot)

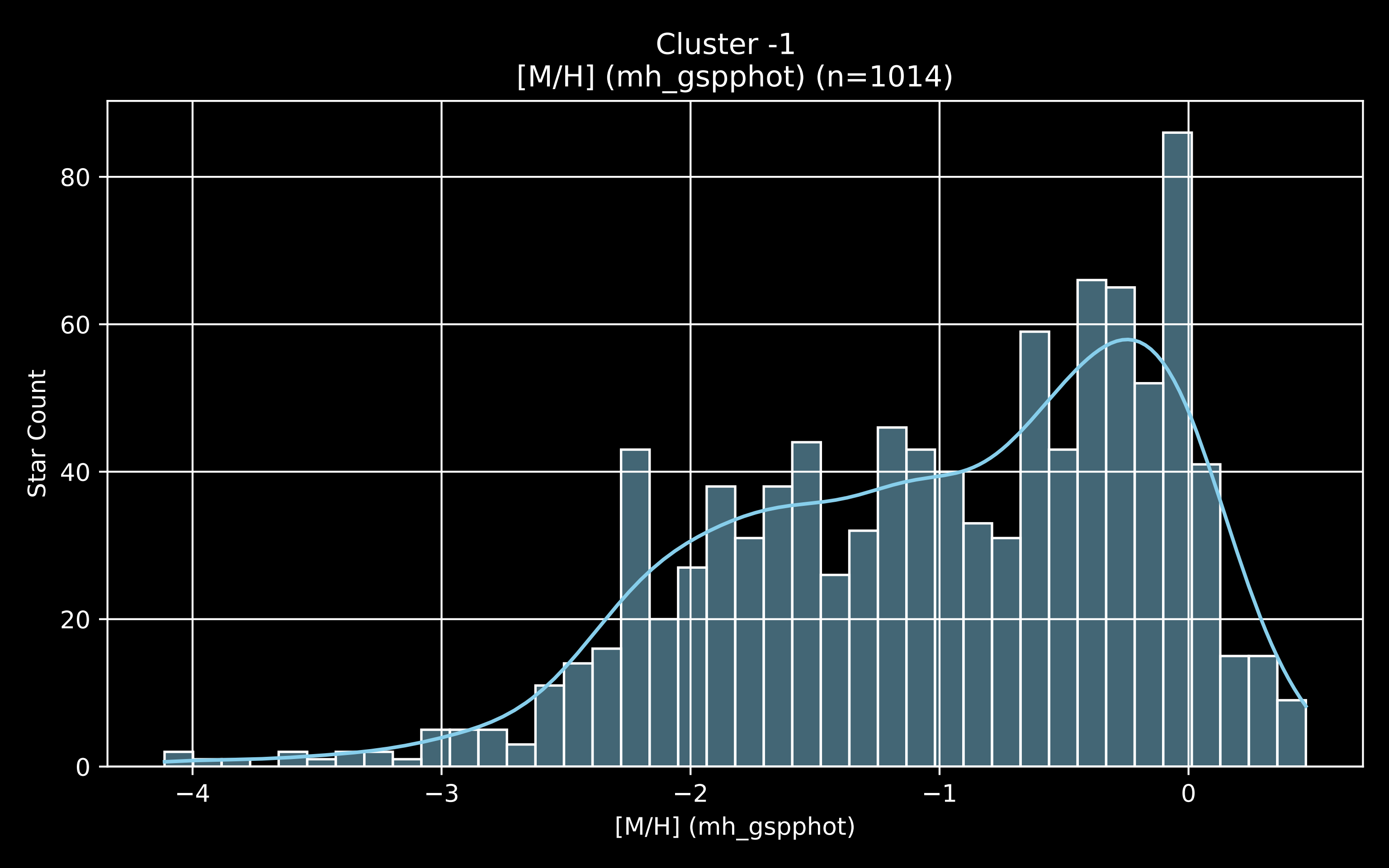


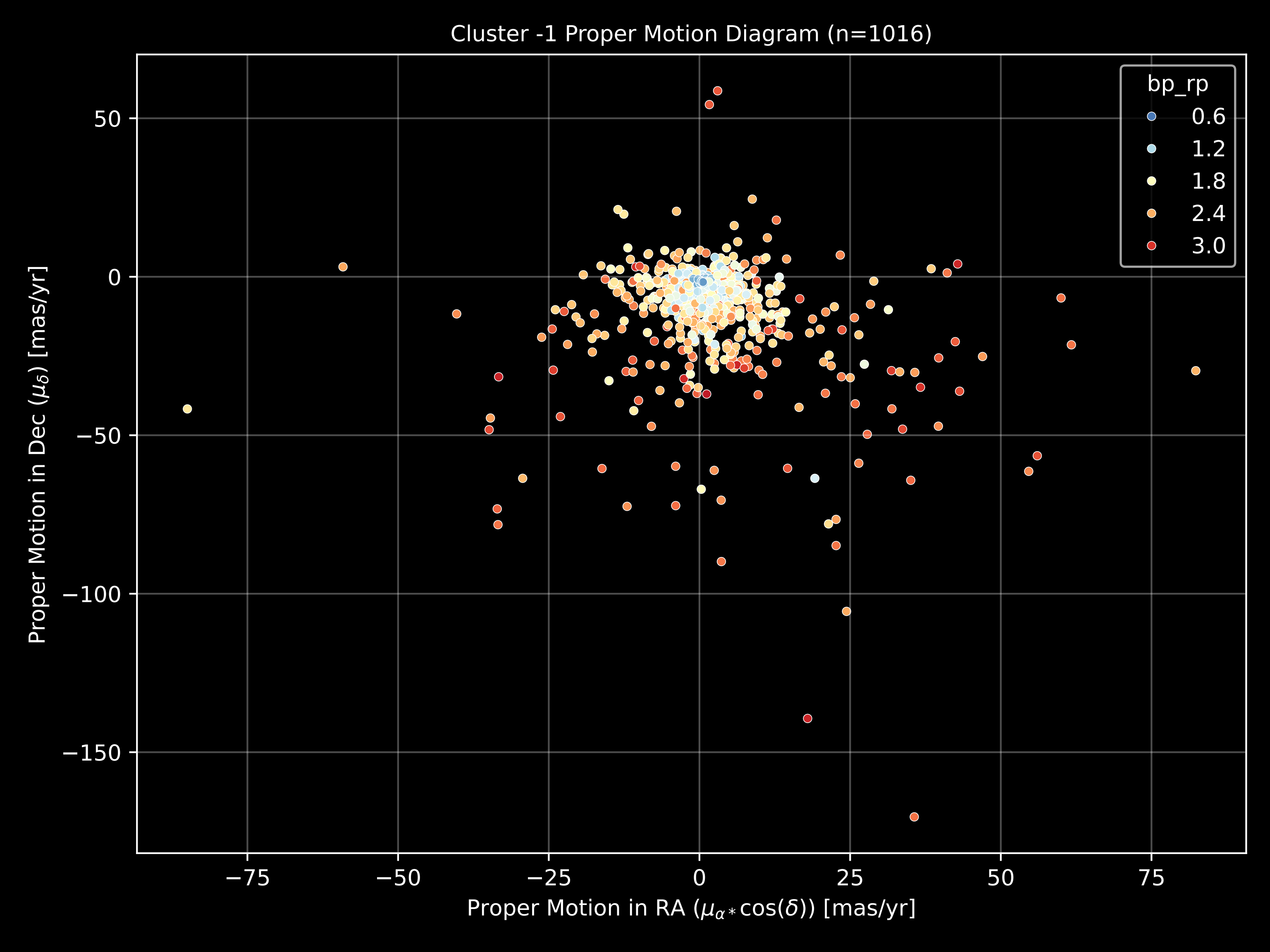
Cluster 105 Proper Motion Diagram (n=18)
bp_rp
0.6
0.8
1.0
1.2
1.4
−0.775
−0.800
−0.825
−0.850
−0.875
−0.900
−0.925
Proper Motion in Dec (μ_δ) [mas/yr]
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

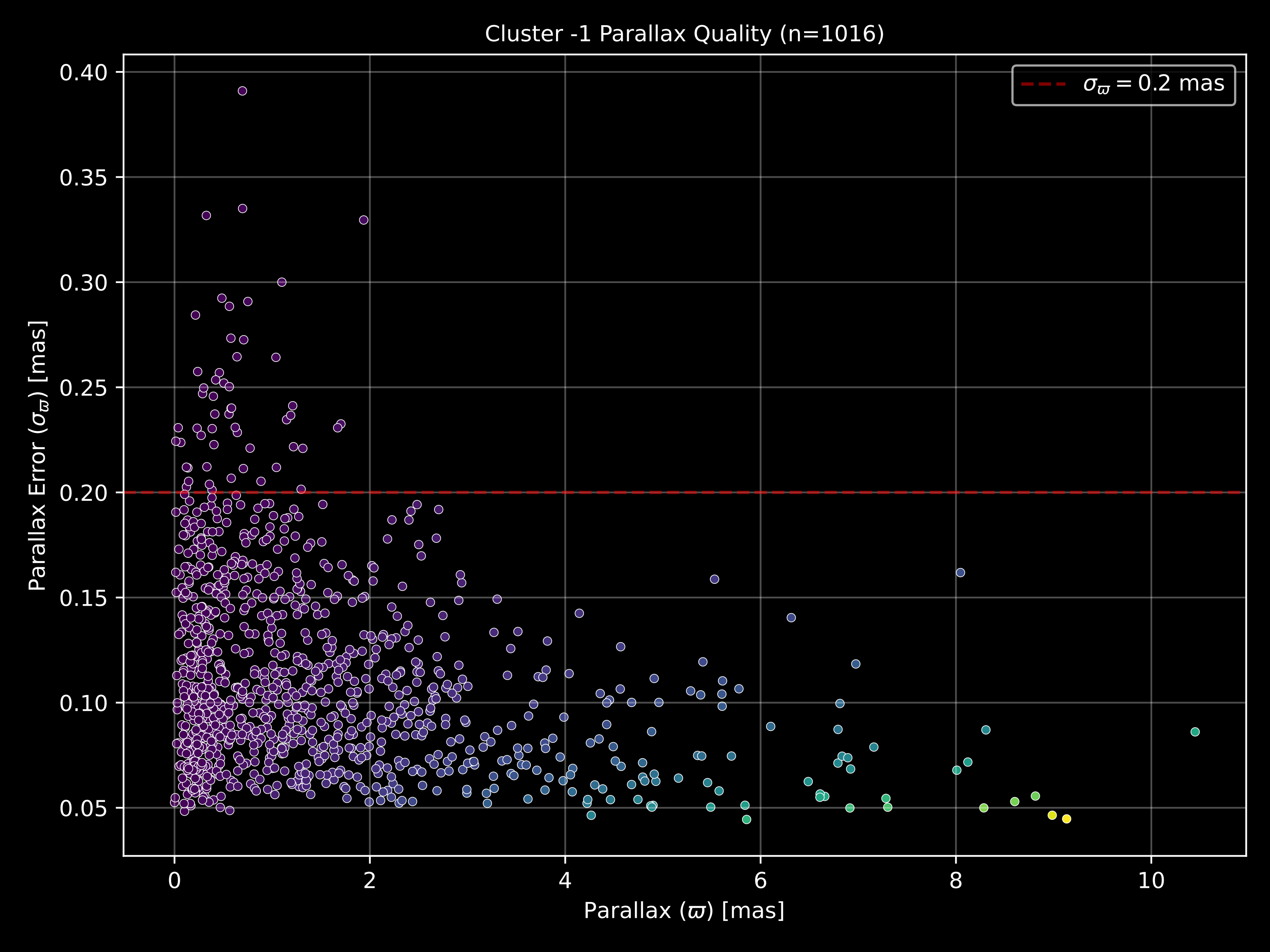

Cluster 105 Parallax Quality (n=18)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.06
0.08
0.10
0.12
0.14
0.16
0.18
0.20
0.22
0.05
0.10
0.15
0.20
0.25
0.30
0.35


LAMOST DR11 Noise Stream Proper Motion Vectors
(n=1016)
Dec (deg)
30
29
28
27
26
95
96
97
98
99
100
RA (deg)

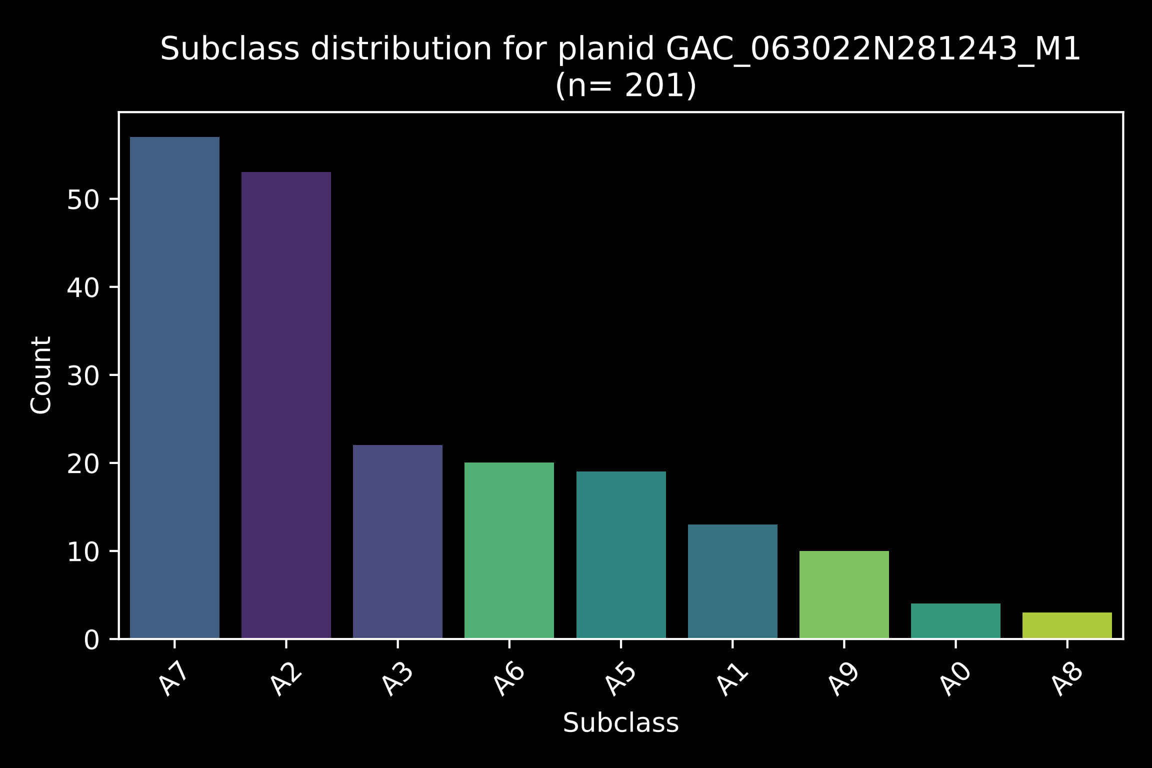

Cluster -1
Count plot of Star Type (n=1016)
Count
1000
800
600
400
200
0
main_sequence
turnoff
red_clump
Star Type

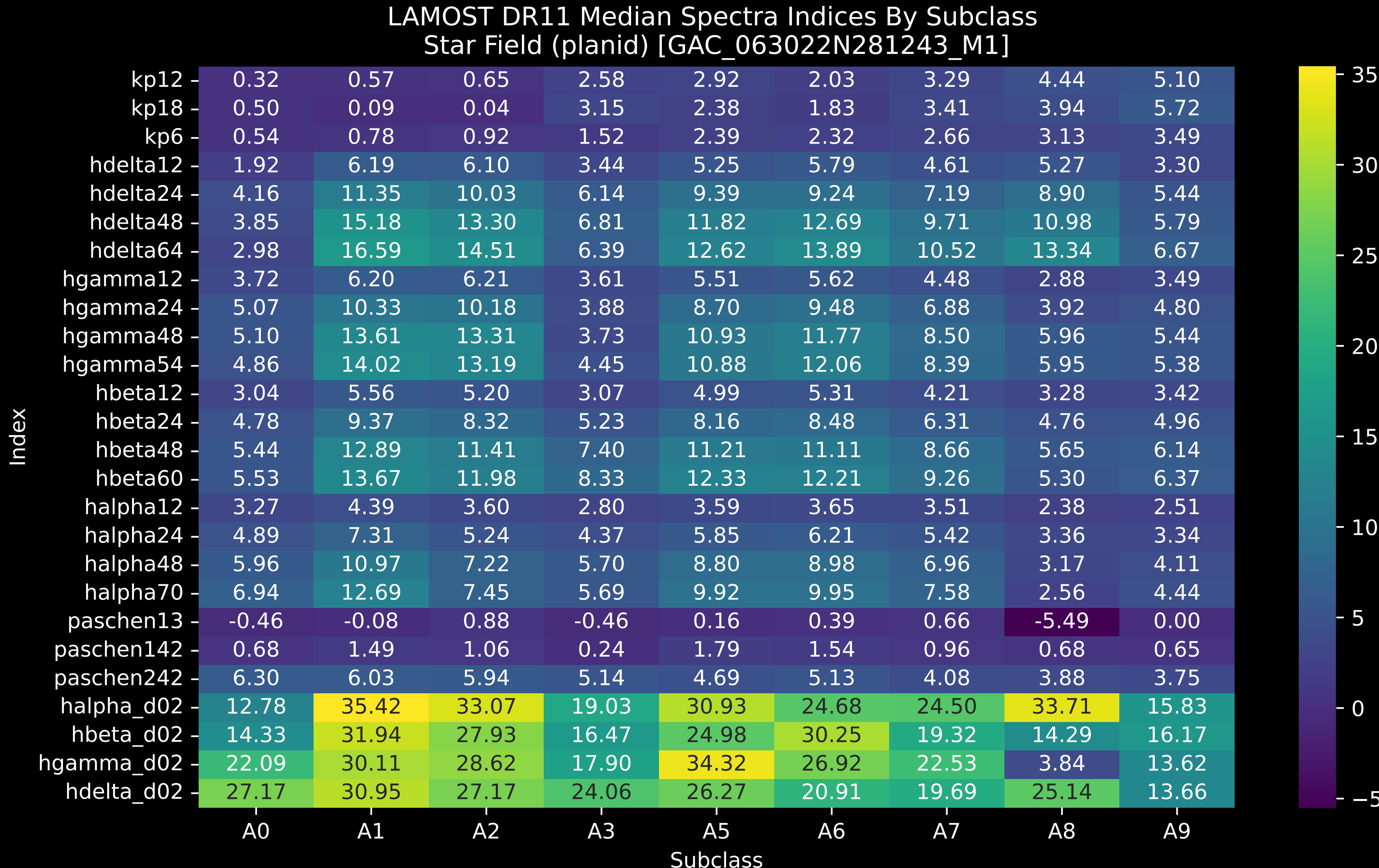

Cluster -1
[M/H] (mh_gspphot) (n=1014)
Star Count
0
20
40
60
80
−4
−3
−2
−1
0
[M/H] (mh_gspphot)

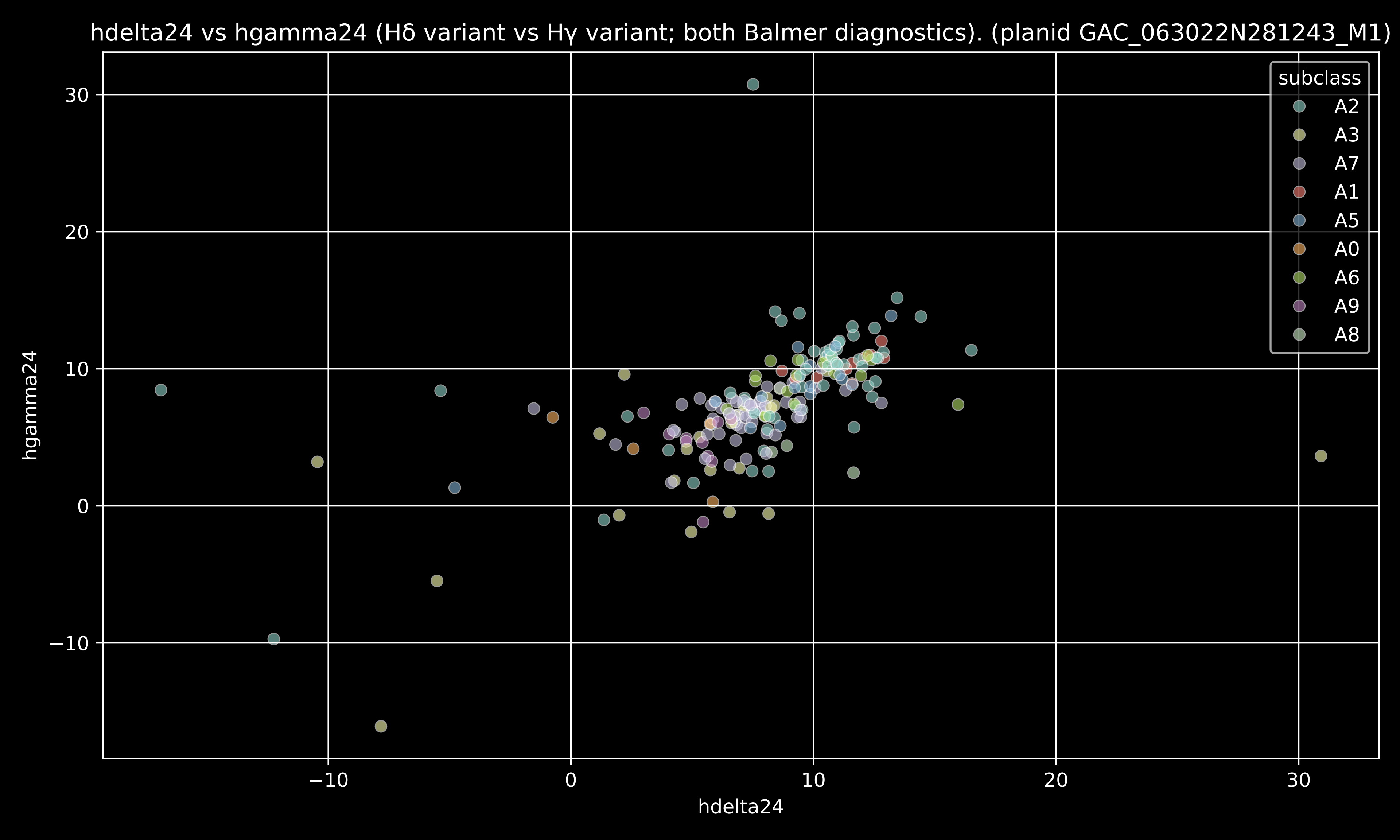

Cluster -1 Proper Motion Diagram (n=1016)
bp_rp
0.6
1.2
1.8
2.4
3.0
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
50
0
−50
−100
−150
−75
−50
−25
0
25
50
75



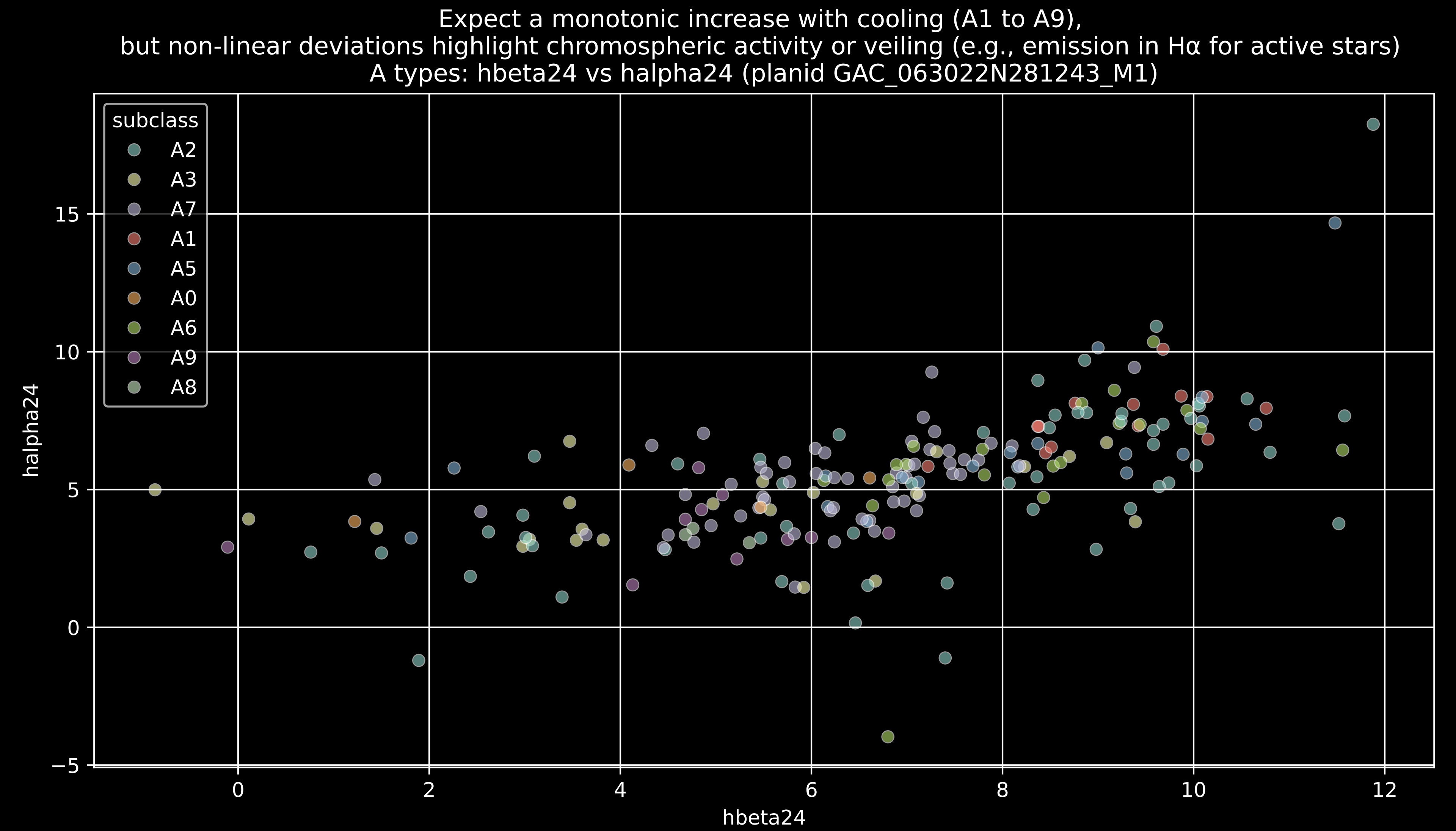
Cluster -1 Parallax Quality (n=1016)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0
2
4
6
8
10


Subclass distribution for planid GAC_063022N281243_M1
(n= 201)
Count
0
10
20
30
40
50
A7
A2
A3
A6
A5
A1
A9
A0
A8
Subclass

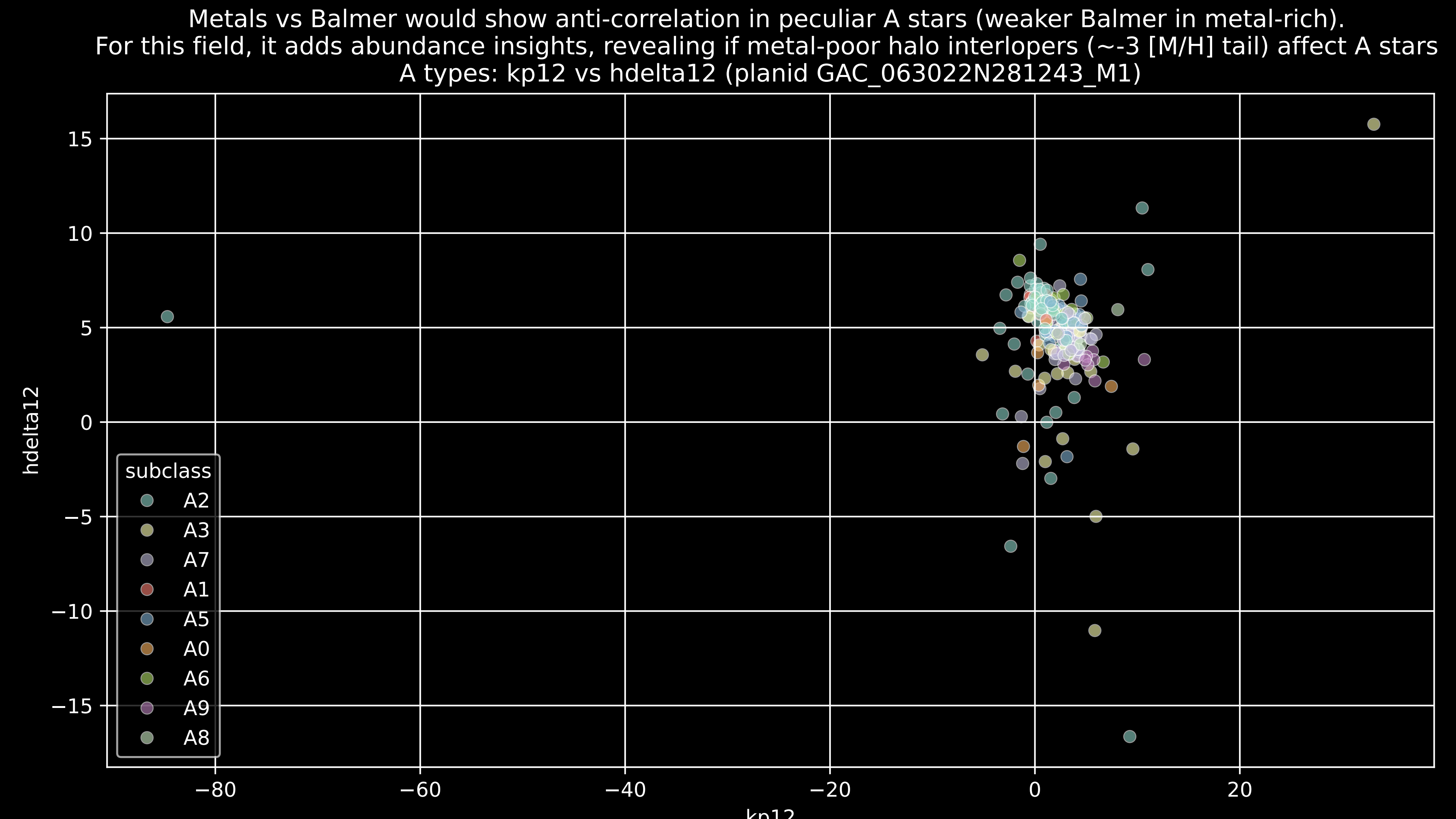

LAMOST DR11 Median Spectra Indices By Subclass
Star Field (planid) [GAC_063022N281243_M1]
Index
Subclass
A0 A1 A2 A3 A5 A6 A7 A8 A9
kp12 0.32 0.57 0.65 2.58 2.92 2.03 3.29 4.44 5.10
kp18 0.50 0.09 0.04 3.15 2.38 1.83 3.41 3.94 5.72
kp6 0.54 0.78 0.92 1.52 2.39 2.32 2.66 3.13 3.49
hdelta12 1.92 6.19 6.10 3.44 5.25 5.79 4.61 5.27 3.30
hdelta24 4.16 11.35 10.03 6.14 9.39 9.24 7.19 8.90 5.44
hdelta48 3.85 15.18 13.30 6.81 11.82 12.69 9.71 10.98 5.79
hdelta64 2.98 16.59 14.51 6.39 12.62 13.89 10.52 13.34 6.67
hgamma12 3.72 6.20 6.21 3.61 5.51 5.62 4.48 2.88 3.49
hgamma24 5.07 10.33 10.18 3.88 8.70 9.48 6.88 3.92 4.80
hgamma48 5.10 13.61 13.31 3.73 10.93 11.77 8.50 5.96 5.44
hgamma54 4.86 14.02 13.19 4.45 10.88 12.06 8.39 5.95 5.38
hbeta12 3.04 5.56 5.20 3.07 4.99 5.31 4.21 3.28 3.42
hbeta24 4.78 9.37 8.32 5.23 8.16 8.48 6.31 4.76 4.96
hbeta48 5.44 12.89 11.41 7.40 11.21 11.11 8.66 5.65 6.14
hbeta60 5.53 13.67 11.98 8.33 12.33 12.21 9.26 5.30 6.37
halpha12 3.27 4.39 3.60 2.80 3.59 3.65 3.51 2.38 2.51
halpha24 4.89 7.31 5.24 4.37 5.85 6.21 5.42 3.36 3.34
halpha48 5.96 10.97 7.22 5.70 8.80 8.98 6.96 3.17 4.11
halpha70 6.94 12.69 7.45 5.69 9.92 9.95 7.58 2.56 4.44
paschen13 -0.46 -0.08 0.88 -0.46 0.16 0.39 0.66 -5.49 0.00
paschen142 0.68 1.49 1.06 0.24 1.79 1.54 0.96 0.68 0.65
paschen242 6.30 6.03 5.94 5.14 4.69 5.13 4.08 3.88 3.75
halpha_d02 12.78 35.42 33.07 19.03 30.93 24.68 24.50 33.71 15.83
hbeta_d02 14.33 31.94 27.93 16.47 24.98 30.25 19.32 14.29 16.17
hgamma_d02 22.09 30.11 28.62 17.90 34.32 26.92 22.53 3.84 13.62
hdelta_d02 27.17 30.95 27.17 24.06 26.27 20.91 19.69 25.14 13.66
35
30
25
20
15
10
5
0
−5



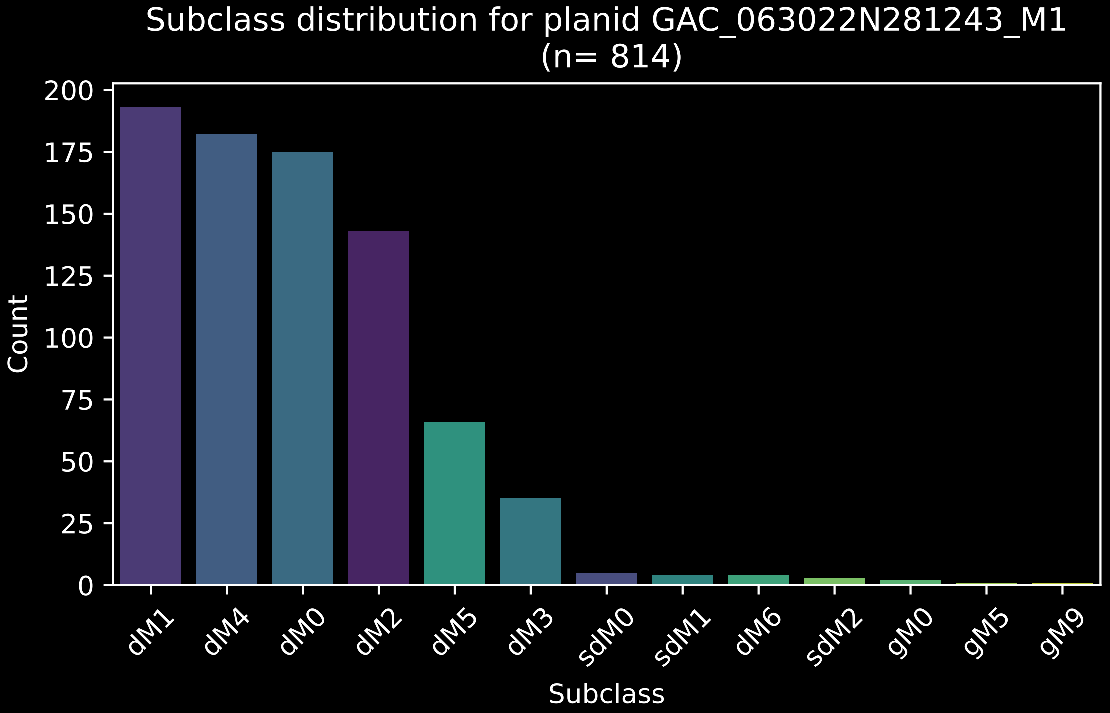
hdelta24 vs hgamma24 (Hδ variant vs Hγ variant; both Balmer diagnostics). (planid GAC_063022N281243_M1)
subclass
A2
A3
A7
A1
A5
A0
A6
A9
A8
30
20
10
0
−10
−10
0
10
20
30
hdelta24
hgamma24



Expect a monotonic increase with cooling (A1 to A9),
but non-linear deviations highlight chromospheric activity or veiling (e.g., emission in Hα for active stars)
A types: hbeta24 vs halpha24 (planid GAC_063022N281243_M1)
subclass
A2
A3
A7
A1
A5
A0
A6
A9
A8
halpha24
hbeta24
15
10
5
0
−5
0
2
4
6
8
10
12

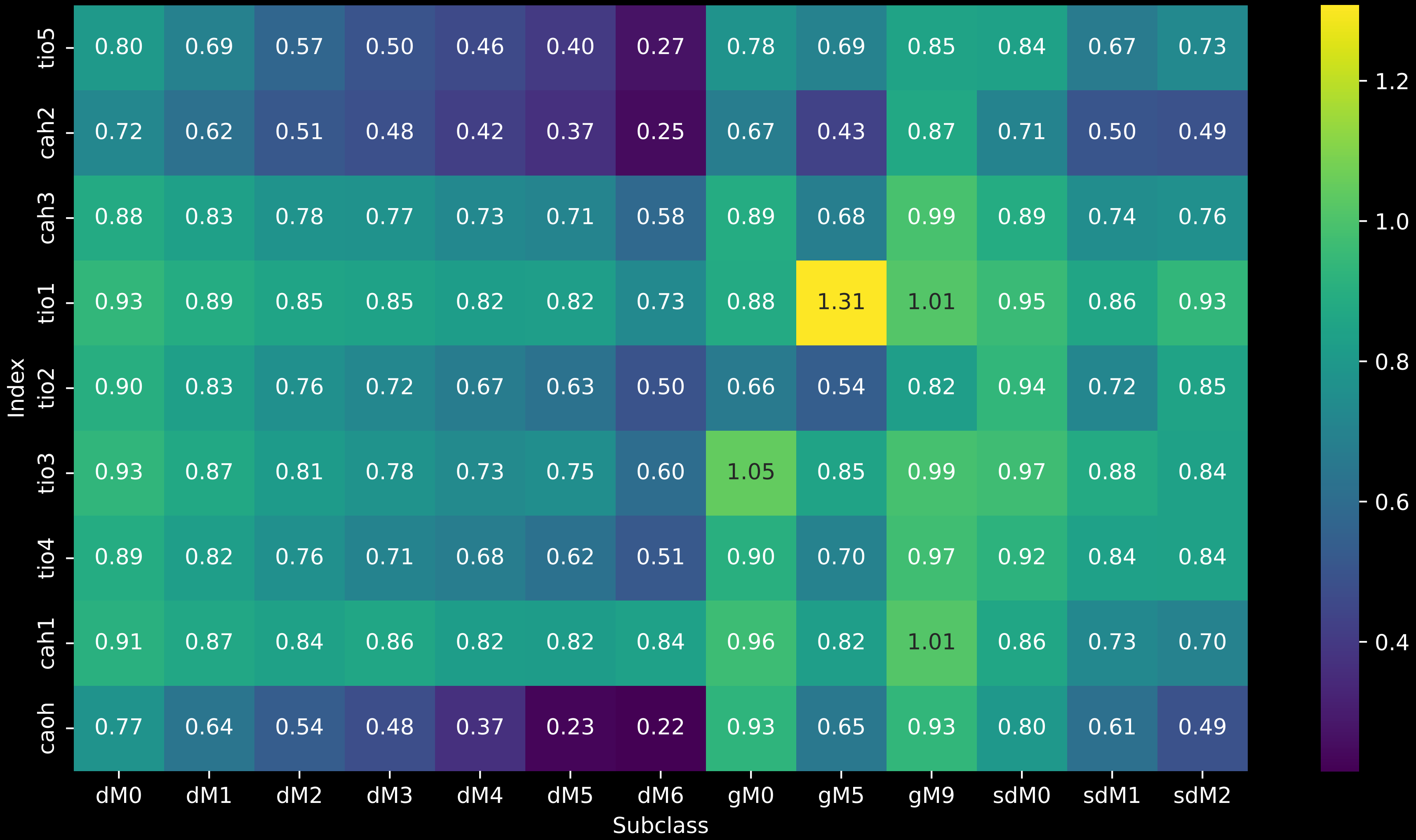


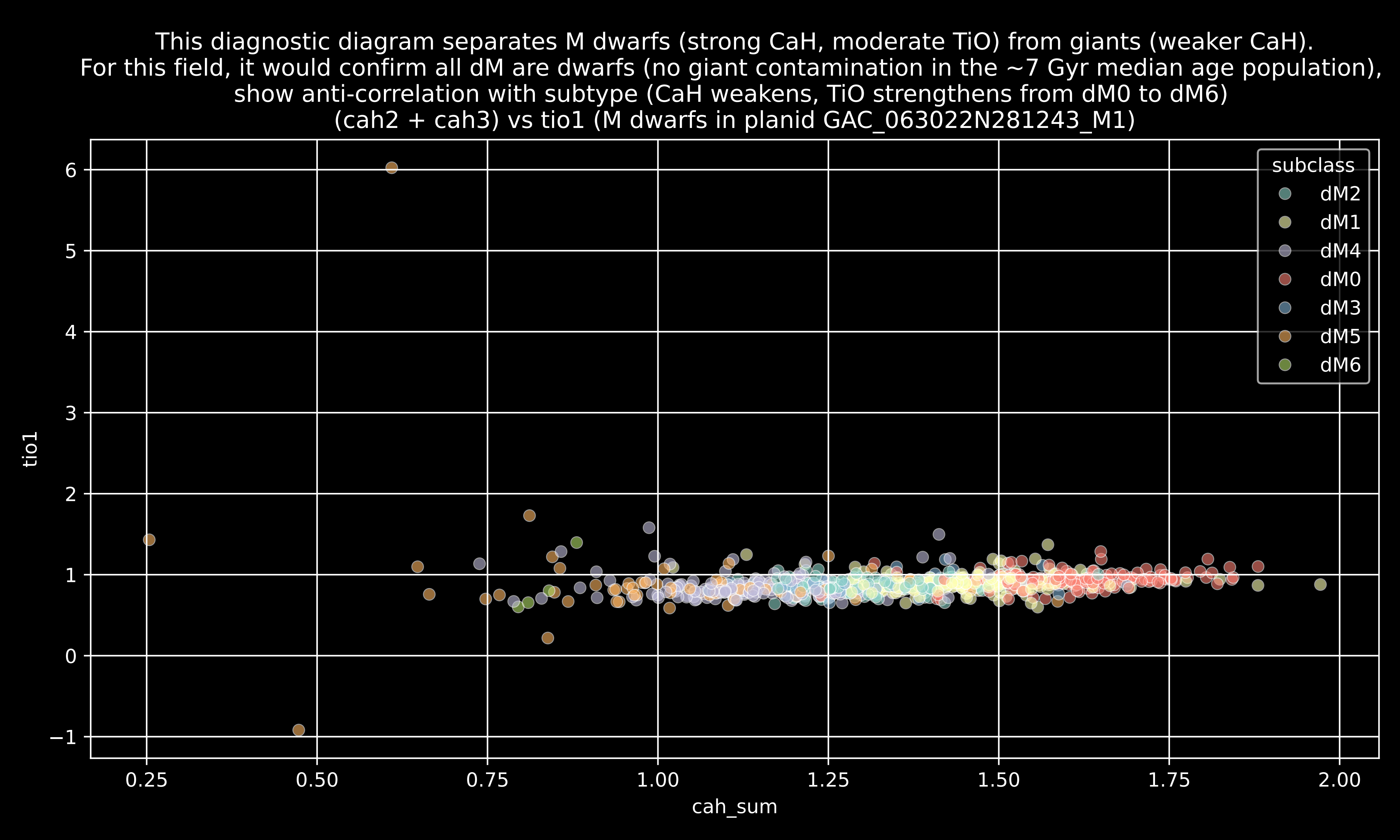
Metals vs Balmer would show anti-correlation in peculiar A stars (weaker Balmer in metal-rich).
For this field, it adds abundance insights, revealing if metal-poor halo interlopers (~-3 [M/H] tail) affect A stars
A types: kp12 vs hdelta12 (planid GAC_063022N281243_M1)
hdelta12
kp12
subclass
A2
A3
A7
A1
A5
A0
A6
A9
A8

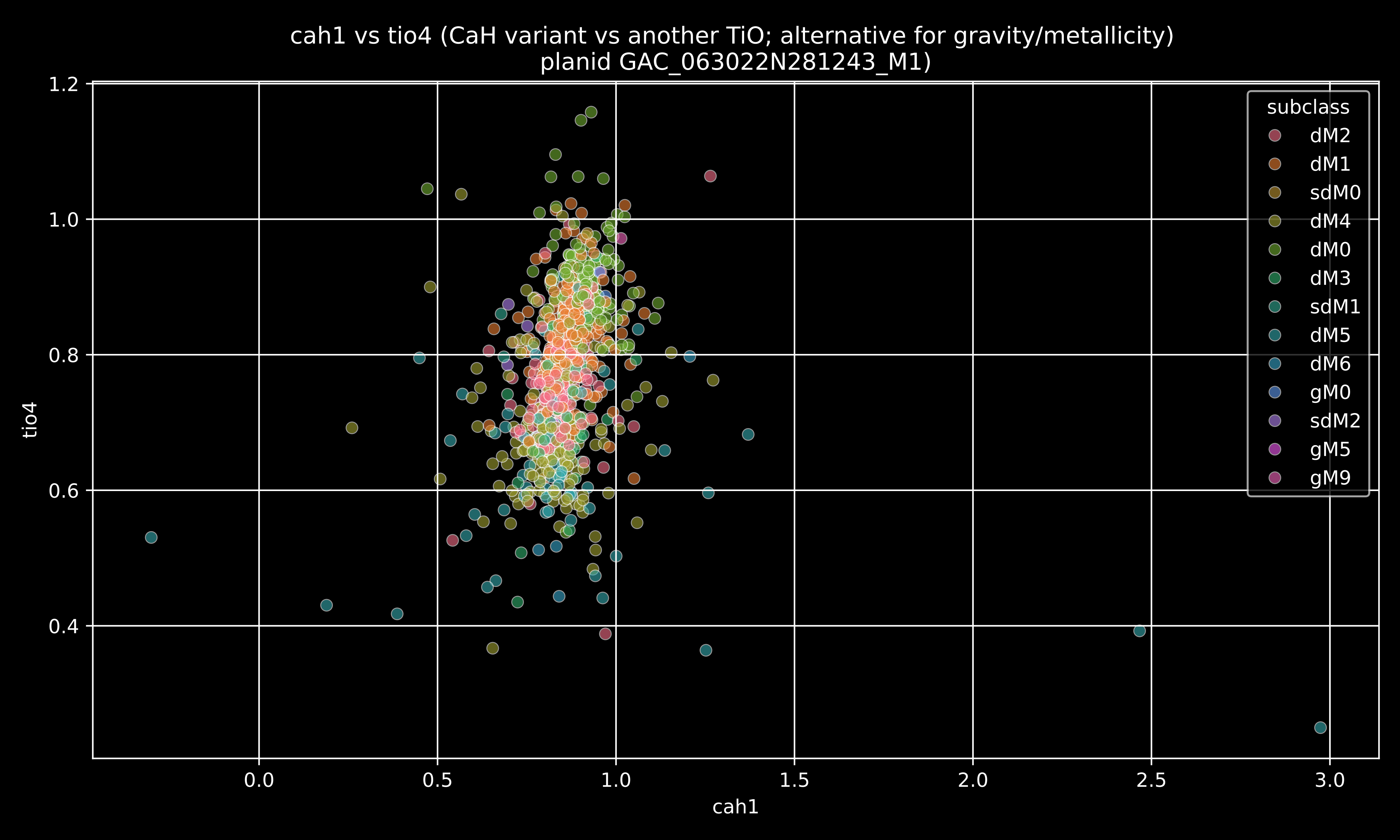


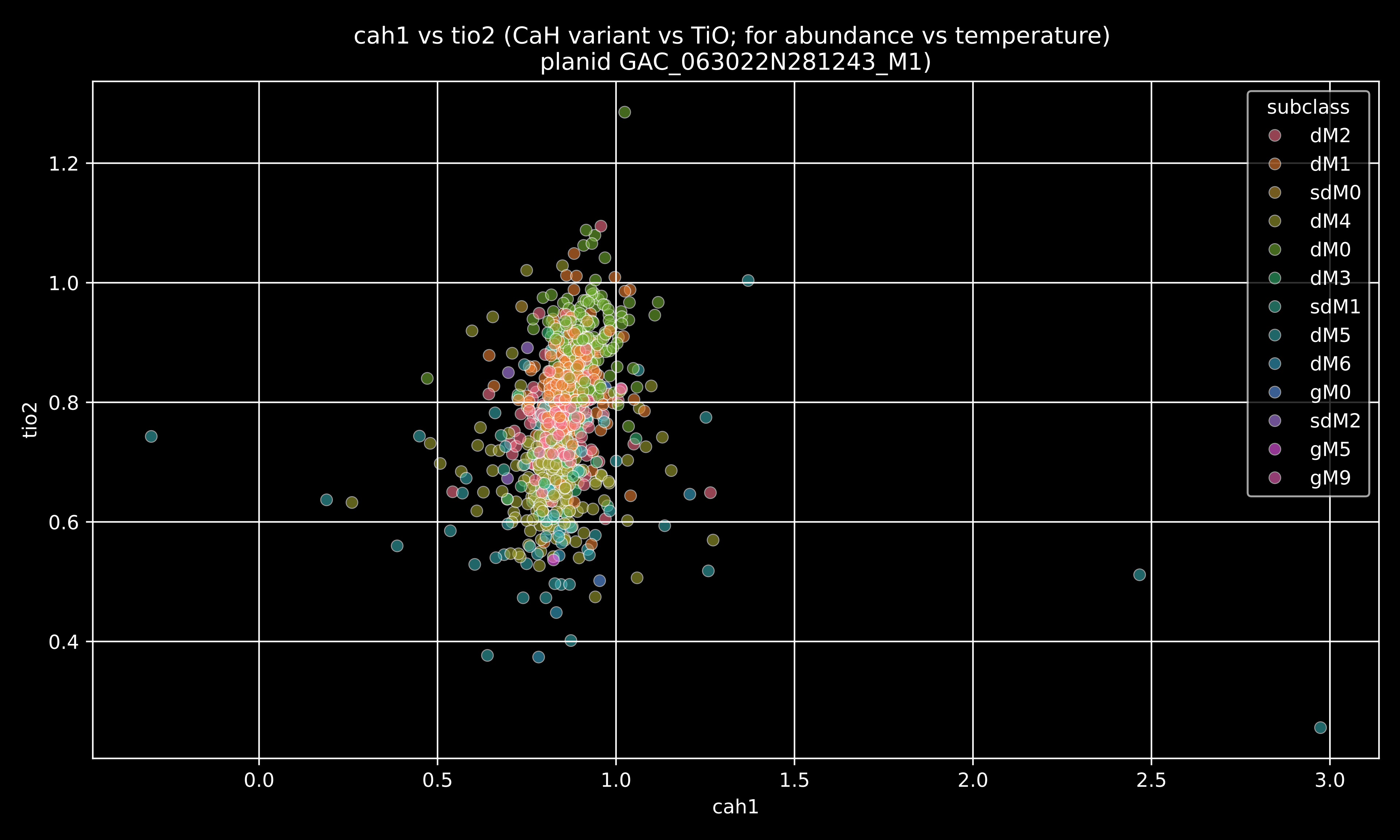
Subclass distribution for planid GAC_063022N281243_M1
(n= 814)
Count
200
175
150
125
100
75
50
25
0
dM1
dM4
dM0
dM2
dM5
dM3
sdM0
sdM1
dM6
sdM2
gM0
gM5
gM9
Subclass

LAMOST DR11 Median Spectra Indices By Subclass
Star Field (planid) [GAC_063022N281243_M1]




This diagnostic diagram separates M dwarfs (strong CaH, moderate TiO) from giants (weaker CaH).
For this field, it would confirm all dM are dwarfs (no giant contamination in the ~7 Gyr median age population),
show anti-correlation with subtype (CaH weakens, TiO strengthens from dM0 to dM6)
(cah2 + cah3) vs tio1 (M dwarfs in planid GAC_063022N281243_M1)
subclass
dM2
dM1
dM4
dM0
dM3
dM5
dM6
tio1
cah_sum
6
5
4
3
2
1
0
−1
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00


cah1 vs tio4 (CaH variant vs another TiO; alternative for gravity/metallicity)
planid GAC_063022N281243_M1)
tio4
cah1
1.2
1.0
0.8
0.6
0.4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
subclass
dM2
dM1
sdM0
dM4
dM0
dM3
sdM1
dM5
dM6
gM0
sdM2
gM5
gM9


cah1 vs tio2 (CaH variant vs TiO; for abundance vs temperature)
planid GAC_063022N281243_M1)
tio2
cah1
1.2
1.0
0.8
0.6
0.4
0.0
0.5
1.0
1.5
2.0
2.5
3.0
subclass
dM2
dM1
sdM0
dM4
dM0
dM3
sdM1
dM5
dM6
gM0
sdM2
gM5
gM9